

Classical Mechanics

Course notes

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1 Survey of the Elementary Principles

1.1 Mechanics of a Particle

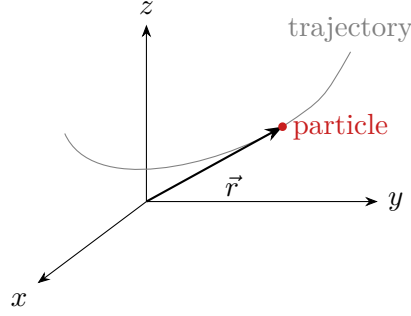


Figure 1.1: Position $\vec{r}$ of a particle moving along its trajectory.

The **velocity** is $\vec{v} := d\vec{r}/dt$, and the **linear momentum** of a particle of mass m is

$$\vec{p} := m\vec{v}.$$

The mechanics of the particle is contained in Newton's second law of motion, which states that there exist reference frames, called **inertial frames**, in which the motion of the particle is described by

$$\vec{F} = \sum_i \vec{F}_i = \frac{d\vec{p}}{dt}.$$

In most cases (constant mass), $\sum_i \vec{F}_i = m d\vec{v}/dt = m d^2\vec{r}/dt^2 =: m\vec{a}$, with $\vec{a}$ the acceleration.

Conservation theorem for the linear momentum. If the total force $\vec{F}$ is zero, then $\dot{\vec{p}} = 0$ and the linear momentum $\vec{p}$ is conserved.

The **angular momentum** of the particle about a point O is

$$\vec{L} := \vec{r} \times \vec{p},$$

and the **moment of force** or **torque** about O is

$$\vec{N} := \vec{r} \times \vec{F}.$$

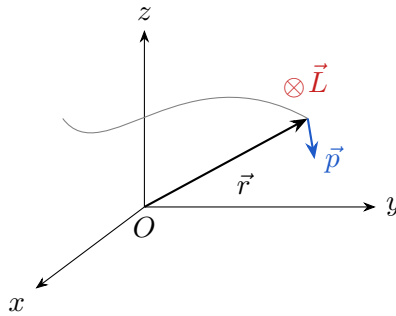


Figure 1.2: Angular momentum $\vec{L} = \vec{r} \times \vec{p}$ about O .

Applying $\vec{r} \times$ to Newton's second law,

$$\vec{N} = \vec{r} \times \vec{F} = \vec{r} \times \frac{d\vec{p}}{dt} = \vec{r} \times \frac{d}{dt}(m\vec{v}) = \frac{d}{dt}(\vec{r} \times \vec{p}) - \underbrace{\vec{v} \times m\vec{v}}_0,$$

so that

$$\vec{N} = \frac{d\vec{L}}{dt}.$$

Conservation theorem for the angular momentum of a particle. If the total torque $\vec{N}$ is zero, then $\dot{\vec{L}} = 0$ and the angular momentum $\vec{L}$ is conserved.

Work and energy. The work done by the external force $\vec{F}$ upon the particle is

$$W_{12} := \int_1^2 \vec{F} \cdot d\vec{l}.$$

Using Newton's second law and assuming constant mass,

$$W_{12} = \int_1^2 \frac{d\vec{p}}{dt} \cdot d\vec{l} = m \int_1^2 \frac{d\vec{v}}{dt} \cdot \frac{d\vec{l}}{dt} dt = m \int_1^2 \vec{v} \cdot \frac{d\vec{v}}{dt} dt = m \int_1^2 \frac{d}{dt} \left(\frac{1}{2} v^2 \right) dt,$$

that is,

$$W_{12} = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2.$$

The scalar quantity $\frac{1}{2} m v^2$ is called the **kinetic energy** and is denoted by T .

If the force is such that **the work is the same for any physically possible path** between points 1 and 2, the force (and the system) is said to be **conservative**: $\oint \vec{F} \cdot d\vec{l} = 0$. From vector analysis, a conservative force can be written as

$$\vec{F} = -\nabla V,$$

where V is called the **potential** or **potential energy**. V is defined up to a constant in space: its zero level is arbitrary. For a conservative force,

$$W_{12} = \int_1^2 \vec{F} \cdot d\vec{l} = - \int_1^2 \nabla V \cdot d\vec{l} = V_1 - V_2.$$

Hence, if there are no dissipative forces,

$$T_1 + V_1 = T_2 + V_2,$$

which is the **energy conservation theorem for a particle**: if the forces acting on a particle are conservative, the total energy $T + V$ is conserved.

If V depends on time, the total energy $T + V$ can still be defined, but it is not conserved (it is as if the zero level of the potential were changing).

1.2 Mechanics of a System of Particles

To generalise the previous results to a system of particles, we must consider both internal and external forces:

$$\sum_j \vec{F}_{ji} + \vec{F}_i^{(e)} = \dot{\vec{p}}_i,$$

where $\vec{F}_{ji}$ is the internal force on particle i due to particle j and $\vec{F}_i^{(e)}$ the external force on i . We assume that the $\vec{F}_{ji}$ satisfy Newton's third law, $\vec{F}_{ij} = -\vec{F}_{ji}$ (**weak law of action and reaction**); this does not hold for all types of forces.

Assuming the mass of each particle does not vary with time,

$$\frac{d^2}{dt^2} \sum_i m_i \vec{r}_i = \sum_i \vec{F}_i^{(e)} + \underbrace{\sum_{\substack{i,j \\ i \neq j}} \vec{F}_{ji}}_{0 \text{ (third law)}}.$$

Defining the **centre of mass**

$$\vec{R} := \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i} = \frac{\sum_i m_i \vec{r}_i}{M},$$

we get

$$M \frac{d^2 \vec{R}}{dt^2} = \vec{F}^{(e)}.$$

The centre of mass moves as if the total external force were acting on the entire mass of the system concentrated at the centre of mass (this holds only because the internal forces cancel, here by the third law).

The total linear momentum of the system is

$$\vec{P} := \sum_i m_i \frac{d\vec{r}_i}{dt} = M \frac{d\vec{R}}{dt}.$$

Conservation theorem for the linear momentum of a system of particles. If the total external force is zero, the total linear momentum is conserved.

The total angular momentum of the system is

$$\vec{L} := \sum_i \vec{r}_i \times \vec{p}_i,$$

and

$$\dot{\vec{L}} = \sum_i \frac{d}{dt} (\vec{r}_i \times \vec{p}_i) = \sum_i \vec{r}_i \times \dot{\vec{p}}_i = \underbrace{\sum_i \vec{r}_i \times \vec{F}_i^{(e)}}_{\vec{N}^{(e)}} + \sum_{\substack{i,j \\ i \neq j}} \vec{r}_i \times \vec{F}_{ji}.$$

Grouping the internal terms in pairs and using the weak law,

$$\vec{r}_i \times \vec{F}_{ji} + \vec{r}_j \times \vec{F}_{ij} = (\vec{r}_i - \vec{r}_j) \times \vec{F}_{ji}.$$

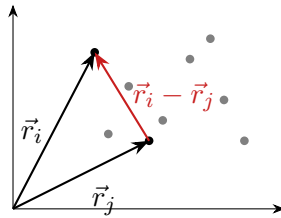


Figure 1.3: The internal torque of a pair vanishes if $\vec{F}_{ji}$ is along $\vec{r}_i - \vec{r}_j$.

If $\vec{F}_{ji}$ lies along the line joining particles i and j , this term vanishes. Together with the weak law this is the **strong law of action and reaction**, and then

$$\frac{d\vec{L}}{dt} = \vec{N}^{(e)}.$$

Conservation theorem for the total angular momentum. $\vec{L}$ is constant in time if the applied (external) torque is zero.

In summary, conservation of linear momentum requires the absence of external forces and the weak law of action and reaction; conservation of angular momentum requires the absence of external torque and the strong law of action and reaction. The strong law of action and reaction is equivalent to having central forces.

There are cases where the weak or strong laws of action and reaction do not hold, and the usual linear or angular momenta are not conserved. In such cases it is usually possible to define a generalised $\vec{P}$ and $\vec{L}$ that are conserved.

Angular momentum about the centre of mass. As shown before, the total linear momentum of the system is the same as if the total mass were concentrated at the centre of mass. For the angular momentum the situation differs.

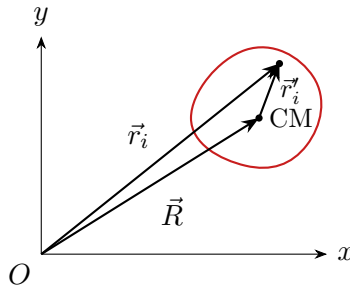


Figure 1.4: Position of particle i relative to the centre of mass.

Writing $\vec{r}_i = \vec{r}'_i + \vec{R}$ and $\vec{v}_i = \vec{v}'_i + \vec{v}$, with $\vec{v}'_i = d\vec{r}'_i/dt$ and $\vec{v} = d\vec{R}/dt$,

$$\begin{aligned}\vec{L} &= \sum_i (\vec{r}'_i + \vec{R}) \times m_i (\vec{v}'_i + \vec{v}) \\ &= \sum_i \vec{r}'_i \times m_i \vec{v}'_i + \left(\sum_i m_i \vec{r}'_i \right) \times \vec{v} + \vec{R} \times \frac{d}{dt} \left(\sum_i m_i \vec{r}'_i \right) + \vec{R} \times M \vec{v}.\end{aligned}$$

Since $\sum_i m_i \vec{r}'_i = 0$ (definition of the centre of mass),

$$\vec{L} = \vec{R} \times M \vec{v} + \sum_i \vec{r}'_i \times \vec{p}'_i.$$

The total angular momentum about a point O is the angular momentum of the motion concentrated at the centre of mass, plus the angular momentum of the motion about the centre of mass.

Energy.

$$W_{12} = \sum_i \int_1^2 \vec{F}_i \cdot d\vec{l}_i = \sum_i \int_1^2 \vec{F}_i^{(e)} \cdot d\vec{l}_i + \sum_{\substack{i,j \\ i \neq j}} \int_1^2 \vec{F}_{ji} \cdot d\vec{l}_i = \sum_i \frac{1}{2} m_i (v_{i,2}^2 - v_{i,1}^2),$$

so

$$W_{12} = T_2 - T_1, \quad T = \sum_i \frac{1}{2} m_i v_i^2.$$

Using the centre-of-mass coordinates,

$$T = \frac{1}{2} \sum_i m_i (\vec{v} + \vec{v}'_i) \cdot (\vec{v} + \vec{v}'_i) = \frac{1}{2} M v^2 + \frac{1}{2} \sum_i m_i v_i'^2 + \underbrace{\vec{v} \cdot \frac{d}{dt} \sum_i m_i \vec{r}'_i}_0,$$

$$T = \frac{1}{2} M v^2 + \frac{1}{2} \sum_i m_i v_i'^2.$$

The kinetic energy consists of two parts: the kinetic energy as if all the mass were concentrated at the centre of mass, plus the kinetic energy of the motion about the centre of mass.

If the external forces are conservative,

$$\sum_i \int_1^2 \vec{F}_i^{(e)} \cdot d\vec{l}_i = - \sum_i \int_1^2 \nabla_i V_i \cdot d\vec{l}_i = - \sum_i V_i \Big|_1^2.$$

If the internal forces are also conservative and satisfy the strong law of action and reaction, the potential of each pair can depend only on the distance,

$$V_{ij} = V_{ij}(|\vec{r}_i - \vec{r}_j|),$$

and then

$$\vec{F}_{ji} = -\nabla_i V_{ij} = +\nabla_j V_{ij} = -\vec{F}_{ij}, \quad \nabla_i V_{ij} = (\vec{r}_i - \vec{r}_j) f$$

for some scalar function f . If V also depended on other pairs of vectors, the force might not lie on the line connecting the particles (only the weak law would hold). The work of the internal forces can be written as

$$\begin{aligned} \sum_{\substack{i,j \\ i \neq j}} \int_1^2 \vec{F}_{ji} \cdot d\vec{l}_i &= -\frac{1}{2} \sum_{i \neq j} \int_1^2 (\nabla_i V_{ij} \cdot d\vec{l}_i + \nabla_j V_{ij} \cdot d\vec{l}_j) = -\frac{1}{2} \sum_{i \neq j} \int_1^2 \nabla_{ij} V_{ij} \cdot d\vec{r}_{ij} \\ &= -\frac{1}{2} \sum_{i \neq j} V_{ij} \Big|_1^2, \end{aligned}$$

where $\vec{r}_{ij} = \vec{r}_i - \vec{r}_j$, ∇_{ij} is the gradient with respect to $\vec{r}_{ij}$, and we used $\nabla_j V_{ij} = -\nabla_i V_{ij}$. Therefore the total potential energy of a conservative system is

$$V = \sum_i V_i + \frac{1}{2} \sum_{i \neq j} V_{ij}.$$

1.3 Constraints

So far, solving the set of differential equations

$$m_i \ddot{\vec{r}}_i = \vec{F}_i^{(e)} + \sum_j \vec{F}_{ji}$$

one can solve several problems, none of which contains a **constraint**. Examples of constraints are a motion forced to occur on a surface, or a rigid body where the distances between particles are fixed.

Classification of constraints.

- **Holonomic**: the conditions of constraint can be written as equations, e.g. $f(\vec{r}_1, \vec{r}_2, \dots; t) = 0$.
- **Non-holonomic**: the conditions of constraint cannot be expressed as equations, e.g. $f(\vec{r}_1, \vec{r}_2, \dots; t) \geq 0$.
- **Rheonomous**: the condition of constraint contains explicit time dependence.
- **Scleronomous**: the condition of constraint does not contain explicit time dependence.

“Imposing constraints on the system is simply another method of stating that there are forces present in the problem that cannot be specified directly but are known rather in terms of their effect on the motion of the system.”

Any constraint reduces the number of independent equations of motion. A system of N particles free from constraints has $3N$ degrees of freedom. If there are k holonomic constraints, we are left with $3N - k$ independent coordinates (**degrees of freedom**). We can then introduce $3N - k$ variables, called **generalised coordinates**, such that

$$\vec{r}_1 = \vec{r}_1(q_1, \dots, q_{3N-k}; t), \quad \dots, \quad \vec{r}_N = \vec{r}_N(q_1, \dots, q_{3N-k}; t),$$

where the transformation is assumed to be invertible. The generalised coordinates q_i are not necessarily vector components; they can even have different physical dimensions.

For non-holonomic constraints it is not possible to reduce the number of independent variables. An example is rolling without slipping, where the condition is not on the coordinates but on the velocities. Non-holonomic constraints must be tackled individually. An integrable differential constraint is a holonomic constraint.

1.4 D'Alembert's Principle and Lagrange's Equations

A **virtual displacement** $\delta\vec{r}_i$ is a change in the configuration of the system resulting from an arbitrary change of the coordinates, consistent with the forces and constraints imposed on the system at the given instant t . It is called virtual to distinguish it from an actual displacement of the system occurring in a time interval dt , during which the forces and constraints may be changing.

For a system in equilibrium, $\vec{F}_i = 0$ and therefore $\sum_i \vec{F}_i \cdot \delta\vec{r}_i = 0$. Splitting the forces into **applied forces** $\vec{F}_i^{(a)}$ and **forces of constraint** $\vec{f}_i$, $\vec{F}_i = \vec{F}_i^{(a)} + \vec{f}_i$,

$$\sum_i \vec{F}_i^{(a)} \cdot \delta\vec{r}_i + \sum_i \vec{f}_i \cdot \delta\vec{r}_i = 0.$$

We restrict ourselves to systems for which the net virtual work of the forces of constraint is zero (e.g. motion restricted to a surface, where the force of constraint is perpendicular to $\delta\vec{r}$). Then we obtain the **principle of virtual work**

$$\sum_i \vec{F}_i^{(a)} \cdot \delta\vec{r}_i = 0.$$

Notice that the $\delta\vec{r}_i$ are no longer independent because of the constraints, so in general $\vec{F}_i^{(a)} \neq 0$.

For a system in motion we write $\vec{F}_i = \dot{\vec{p}}_i$ as $\vec{F}_i - \dot{\vec{p}}_i = 0$: the system is in equilibrium under the actual force plus a “reversed effective force” $-\dot{\vec{p}}_i$. Then

$$\sum_i (\vec{F}_i - \dot{\vec{p}}_i) \cdot \delta\vec{r}_i = 0 \quad \implies \quad \sum_i (\vec{F}_i^{(a)} - \dot{\vec{p}}_i) \cdot \delta\vec{r}_i + \underbrace{\sum_i \vec{f}_i \cdot \delta\vec{r}_i}_0 = 0,$$

which is **D'Alembert's principle**:

$$\sum_i (\vec{F}_i^{(a)} - \dot{\vec{p}}_i) \cdot \delta \vec{r}_i = 0. \quad (1.1)$$

Let us now transform this equation to the generalised (and independent) coordinates q_j . From here on we use Einstein's summation convention. With holonomic constraints,

$$\delta \vec{r}_i = \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j, \quad \vec{v}_i = \frac{d\vec{r}_i}{dt} = \frac{\partial \vec{r}_i}{\partial q_k} \dot{q}_k + \frac{\partial \vec{r}_i}{\partial t} \implies \frac{\partial \vec{v}_i}{\partial \dot{q}_j} = \frac{\partial \vec{r}_i}{\partial q_j}.$$

Term by term:

$$\vec{F}_i^{(a)} \cdot \delta \vec{r}_i = \vec{F}_i^{(a)} \cdot \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j =: Q_j \delta q_j, \quad Q_j := \vec{F}_i^{(a)} \cdot \frac{\partial \vec{r}_i}{\partial q_j},$$

where Q_j is the **generalised force** (which need not have dimensions of force). For the other term,

$$\sum_i \dot{\vec{p}}_i \cdot \delta \vec{r}_i = \sum_i m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j = \sum_i \left[\frac{d}{dt} \left(m_i \dot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \right) - m_i \dot{\vec{r}}_i \cdot \frac{d}{dt} \left(\frac{\partial \vec{r}_i}{\partial q_j} \right) \right] \delta q_j.$$

Since

$$\frac{d}{dt} \left(\frac{\partial \vec{r}_i}{\partial q_j} \right) = \frac{\partial^2 \vec{r}_i}{\partial q_j \partial q_k} \dot{q}_k + \frac{\partial}{\partial t} \left(\frac{\partial \vec{r}_i}{\partial q_j} \right) = \frac{\partial}{\partial q_j} \left[\frac{\partial \vec{r}_i}{\partial q_k} \dot{q}_k + \frac{\partial \vec{r}_i}{\partial t} \right] = \frac{\partial \vec{v}_i}{\partial \dot{q}_j},$$

we get

$$m_i \dot{\vec{r}}_i \cdot \frac{d}{dt} \left(\frac{\partial \vec{r}_i}{\partial q_j} \right) = m_i \vec{v}_i \cdot \frac{\partial \vec{v}_i}{\partial \dot{q}_j} = \frac{\partial}{\partial q_j} \left(\frac{1}{2} m_i v_i^2 \right), \quad m_i \dot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} = m_i \vec{v}_i \cdot \frac{\partial \vec{v}_i}{\partial \dot{q}_j} = \frac{\partial}{\partial \dot{q}_j} \left(\frac{1}{2} m_i v_i^2 \right).$$

Thus (1.1) in generalised coordinates reads

$$\left[Q_j - \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) + \frac{\partial T}{\partial q_j} \right] \delta q_j = 0,$$

and, since the δq_j are independent,

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j.$$

If the applied forces are conservative,

$$Q_j = \vec{F}_i^{(a)} \cdot \frac{\partial \vec{r}_i}{\partial q_j} = -\nabla_i V \cdot \frac{\partial \vec{r}_i}{\partial q_j} = -\frac{\partial V}{\partial q_j} \implies \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial}{\partial q_j} (T - V) = 0.$$

Since in this case V does not depend on the $\dot{q}_j$,

$$\frac{d}{dt} \left(\frac{\partial (T - V)}{\partial \dot{q}_j} \right) - \frac{\partial (T - V)}{\partial q_j} = 0.$$

Defining the **Lagrangian** $L := T - V$, we obtain **Lagrange's equations**

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = 0.$$

The Lagrangian is not unique: for instance $L' = L + df(q, t)/dt$ gives the same equations of motion.

2 Variational Principles and Lagrange's Equations

2.1 Hamilton's Principle

The derivation of Lagrange's equations in the last chapter was from a differential principle. However, it is also possible to derive them from an integral principle.

The instantaneous configuration of a system is described by the values of the n generalised coordinates $q_1, \dots, q_n$, and corresponds to a particular point in a Cartesian hyperspace where the q 's form the n coordinate axes. This n -dimensional space is known as **configuration space**. The "motion of the system" refers to the motion of the system point along a path in configuration space, time being a parameter of the path.

The integral Hamilton's principle describes the motion of those mechanical systems for which **all forces (except the forces of constraint) are derivable from a generalised scalar potential** that may be a function of the coordinates, velocities and time. Such systems are called **monogenic**. If $V = V(\vec{r})$, the monogenic system is also conservative.

For monogenic systems, **Hamilton's principle** states: the motion of the system from time t_1 to time t_2 is such that the line integral (called the **action** or action integral)

$$I = \int_{t_1}^{t_2} dt L, \quad L = T - V,$$

has a stationary value for the actual path of the motion:

$$\delta I = \delta \int_{t_1}^{t_2} dt L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t) = 0.$$

When the system constraints are holonomic, Hamilton's principle is both a necessary and sufficient condition for Lagrange's equations.

2.3 Derivation of Lagrange's Equations from Hamilton's Principle

With $\delta \dot{q} = d(\delta q)/dt$,

$$\begin{aligned} \delta I &= \delta \int_{t_1}^{t_2} dt L(q, \dot{q}) = \int_{t_1}^{t_2} dt \left[\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] \\ &= \int_{t_1}^{t_2} dt \left[\frac{\partial L}{\partial q} \delta q + \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \delta q \right) - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q \right] \\ &= \underbrace{\frac{\partial L}{\partial \dot{q}} \delta q \Big|_{t_1}^{t_2}}_0 + \int_{t_1}^{t_2} dt \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \delta q, \end{aligned}$$

since $\delta q(t_1) = \delta q(t_2) = 0$. The variation δq is otherwise arbitrary, so by the fundamental lemma of the calculus of variations, $\delta I = 0$ implies

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = 0.$$

2.4 Extension of Hamilton's Principle to Non-Holonomic Systems

The variational principle cannot be used directly when non-holonomic constraints are present, since then the δq are not independent. However, when the constraint equations can be written as

$$f_\alpha(q_i, \dot{q}_i) = 0,$$

the system is called **semi-holonomic**, and the variational principle can be used by considering $\sum_{\alpha} \lambda_{\alpha} f_{\alpha} = 0$ with **Lagrange multipliers** $\lambda_{\alpha} = \lambda_{\alpha}(t)$:

$$I = \int_{t_1}^{t_2} dt \left(L + \sum_{\alpha} \lambda_{\alpha} f_{\alpha} \right).$$

The equations of motion are

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) - \frac{\partial L}{\partial q_k} = Q_k, \quad Q_k = \sum_{\alpha=1}^m \lambda_{\alpha} \left[\frac{\partial f_{\alpha}}{\partial q_k} - \frac{d}{dt} \left(\frac{\partial f_{\alpha}}{\partial \dot{q}_k} \right) \right] - \frac{d\lambda_{\alpha}}{dt} \frac{\partial f_{\alpha}}{\partial \dot{q}_k}.$$

This method can also be used for holonomic systems to compute the forces of constraint.

2.6 Conservation Theorems and Symmetry Properties

Even when complete solutions cannot be obtained, it is often possible to extract a large amount of information about the physical nature of the motion. **First integrals**, $f(q_i, \dot{q}_i, t) = \text{const}$, tell us something physical about the system.

The **generalised momentum** conjugate to q_i is

$$p_i := \frac{\partial L}{\partial \dot{q}_i}.$$

If the Lagrangian does not depend on a generalised coordinate q_i , that coordinate is called **cyclic** or **ignorable**. Then Lagrange's equation becomes

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = 0 \quad \implies \quad p_i = \frac{\partial L}{\partial \dot{q}_i} = \text{const} :$$

the generalised momentum conjugate to a cyclic coordinate is conserved. This assumes that the coordinates are all independent of each other. The momentum conservation theorems are closely connected with the **symmetry properties** of the system.

2.7 Energy Function and the Conservation of Energy

Consider the total time derivative of $L = L(q, \dot{q}, t)$:

$$\frac{dL}{dt} = \frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \frac{\partial L}{\partial t} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \frac{\partial L}{\partial t} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right) + \frac{\partial L}{\partial t},$$

using Lagrange's equations. Hence

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \dot{q}_i - L \right) + \frac{\partial L}{\partial t} = 0.$$

Defining the **energy function**

$$h(q, \dot{q}, t) := \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i - L, \quad \frac{dh}{dt} = - \frac{\partial L}{\partial t}.$$

Thus, if the Lagrangian is not an explicit function of time, h is conserved: $h = \text{const}$ is a first integral of the motion, called **Jacobi's integral**.

Sometimes h coincides with the energy of the system and sometimes it does not. The kinetic energy can be written as $T = T_0 + T_1 + T_2$, where $T_0 = T_0(q, t)$, $T_1 = T_1(q, \dot{q}, t)$ is linear in

the generalised velocities and $T_2 = T_2(q, \dot{q}, t)$ is quadratic in them. The Lagrangian can be decomposed similarly,

$$L(q, \dot{q}, t) = L_0(q, t) + L_1(q, \dot{q}, t) + L_2(q, \dot{q}, t),$$

where L_n is a **homogeneous function** of degree n in the $\dot{q}$'s, $f(\alpha x) = \alpha^n f(x)$. There is no reason to restrict the Lagrangian to this form, but it is fulfilled by most systems. By **Euler's theorem**, if f is homogeneous of degree n , then $\sum_i x_i \partial f / \partial x_i = n f$. Thus

$$h = 2L_2 + L_1 - L = L_2 - L_0.$$

If $V \neq V(\dot{q})$ and the coordinates do not depend explicitly on time ($T = T_2$), then

$$h = T + V = E :$$

the energy function is indeed the total energy.

If there are frictional forces derivable from a **dissipation function** $\mathcal{F}$, Lagrange's equations become

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} + \frac{\partial \mathcal{F}}{\partial \dot{q}_i} = 0,$$

and

$$\frac{dh}{dt} + \frac{\partial L}{\partial t} = - \frac{\partial \mathcal{F}}{\partial \dot{q}_i} \dot{q}_i.$$

By definition $\mathcal{F}$ is a homogeneous function of degree 2 in the $\dot{q}$'s, so

$$\frac{dh}{dt} + \frac{\partial L}{\partial t} = -2\mathcal{F}.$$

If L does not depend explicitly on time, $dh/dt = -2\mathcal{F}$, which is the rate of energy dissipation.

3 The Central Force Problem

3.1 Reduction to the Equivalent One-Body Problem

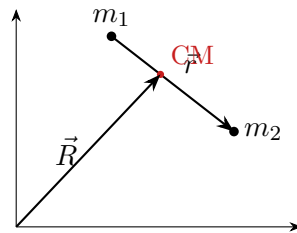


Figure 3.1: Two bodies: centre of mass $\vec{R}$ and relative position $\vec{r} = \vec{r}_2 - \vec{r}_1$.

A monogenic system of two particles has six degrees of freedom, which we take as the centre of mass $\vec{R}$ and the relative vector $\vec{r} = \vec{r}_2 - \vec{r}_1$. The Lagrangian is

$$L = T(\dot{\vec{R}}, \dot{\vec{r}}) - U(\vec{r}, \dot{\vec{r}}, \dots),$$

with

$$T = \underbrace{\frac{1}{2}(m_1 + m_2)\dot{\vec{R}}^2}_{\text{centre of mass}} + \underbrace{\frac{1}{2}m_1\dot{\vec{r}}_1^2 + \frac{1}{2}m_2\dot{\vec{r}}_2^2}_{\text{motion relative to the CM}}.$$

Since $\vec{r}_1 = \vec{R} + \vec{r}'_1$, $\vec{r}_2 = \vec{R} + \vec{r}'_2$ and $\vec{R} = (m_1\vec{r}_1 + m_2\vec{r}_2)/(m_1 + m_2)$,

$$\vec{r}'_1 = -\frac{m_2}{m_1 + m_2}\vec{r}, \quad \vec{r}'_2 = \frac{m_1}{m_1 + m_2}\vec{r},$$

and

$$L = \frac{1}{2}(m_1 + m_2)\dot{\vec{R}}^2 + \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}\dot{\vec{r}}^2 - U(\vec{r}, \dot{\vec{r}}, \dots).$$

$\vec{R}$ is cyclic, so the centre of mass is either at rest or moving uniformly, and that term can be dropped. The rest of the Lagrangian is exactly what we would have for a fixed centre of force with a single particle at distance $\vec{r}$ from it, with the **reduced mass**

$$\mu = \frac{m_1m_2}{m_1 + m_2}, \quad \frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}.$$

Thus the central-force motion of two bodies about their centre of mass can always be reduced to an equivalent one-body problem.

3.2 The Equations of Motion and First Integrals

We restrict the problem to conservative central forces, with potential $V = V(r)$, so that the force is always along $\vec{r}$. The problem has spherical symmetry: no rotation about any axis can alter the result, so the total angular momentum $\vec{L} = \vec{r} \times \vec{p}$ must be conserved. Since $\vec{r} \perp \vec{L}$, $\vec{r}$ must lie in a plane whose normal is parallel to $\vec{L}$ (if $\vec{L} = 0$, the motion is along a line through the centre of force). Since the motion occurs in a plane, we fix one of the spherical coordinates, $\varphi = \pi/2$ (measuring φ from the z axis, normal to the plane of motion, and θ as the angle in the plane).

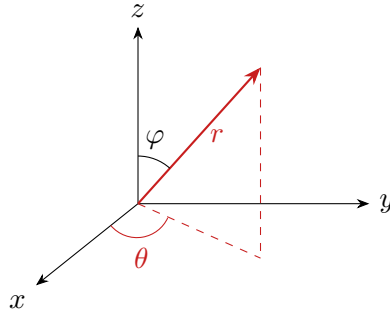


Figure 3.2: Spherical coordinates; for planar motion $\varphi = \pi/2$ is fixed.

Two of the conserved quantities (the direction of $\vec{L}$) have been used to fix the plane of motion, so we go from three to two degrees of freedom; the magnitude of $\vec{L}$ is still left. Then

$$L = T - V = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r).$$

θ is cyclic, so its conjugate momentum, the angular momentum, is conserved:

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}, \quad \frac{d}{dt}p_\theta = \frac{d}{dt}(mr^2\dot{\theta}) = 0,$$

giving the first integral

$$mr^2\dot{\theta} = l,$$

the constant magnitude of the angular momentum. Notice that

$$\frac{d}{dt}(mr^2\dot{\theta}) = 0 \iff \frac{d}{dt}\left(\frac{1}{2}r^2\dot{\theta}\right) = 0,$$

which is the **conservation of areal velocity** (Kepler's second law); it applies to any central force problem with $V = V(r)$.

The Lagrange equation for r is

$$\frac{d}{dt}(m\dot{r}) - mr\dot{\theta}^2 + \frac{\partial V}{\partial r} = 0 \quad \Longrightarrow \quad m\ddot{r} - \frac{l^2}{mr^3} = -\frac{\partial V}{\partial r},$$

using $mr^2\dot{\theta} = l$. There is another first integral, the total energy

$$E = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + V(r),$$

which can be derived from the equation of motion:

$$\begin{aligned} m\ddot{r} &= -\frac{d}{dr} \left(V + \frac{1}{2} \frac{l^2}{mr^2} \right) \quad \Longrightarrow \quad m\ddot{r}\dot{r} = -\frac{d}{dt} \left(V + \frac{1}{2} \frac{l^2}{mr^2} \right) \\ &\Longrightarrow \quad \frac{d}{dt} \left(\frac{1}{2} m\dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2 + V(r) \right) = 0, \end{aligned}$$

where $V + l^2/2mr^2$ is the **effective potential**.

There are only two equations left to solve. From the energy,

$$t = \int_{r_0}^r \frac{dr}{\sqrt{\frac{2}{m} \left(E - V - \frac{l^2}{2mr^2} \right)}},$$

and then

$$\theta = l \int_0^t \frac{dt}{mr^2} + \theta_0.$$

The problem is reduced to “quadratures”, which can be evaluated by numerical integration if necessary.

3.3 The Equivalent One-Dimensional Problem, and Classification of Orbits

Assume a potential of the form $V = -k/r$. The effective potential is

$$V_{\text{eff}} = -\frac{k}{r} + \frac{l^2}{2mr^2},$$

and from energy conservation, $E = \frac{1}{2}m\dot{r}^2 + V_{\text{eff}} \geq V_{\text{eff}}$.

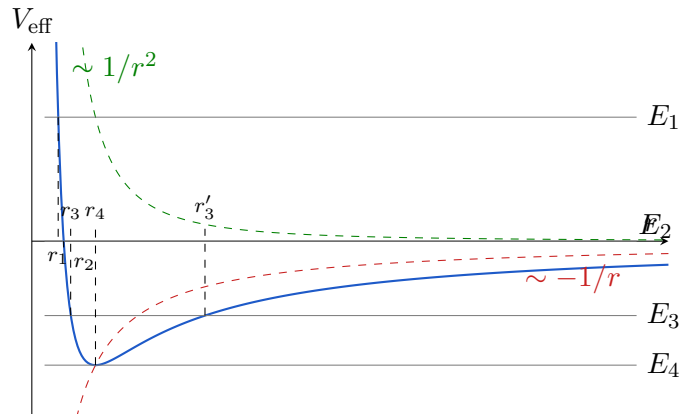


Figure 3.3: Effective potential for $V = -k/r$ and the four types of orbits.

Thus

- $E_1 > 0$: **unbounded orbit** with minimum radius r_1 (hyperbola);
- $E_2 = 0$: unbounded orbit with minimum radius r_2 (parabola);
- $E_3 < 0$: **bounded** (but not necessarily closed) orbit between r_3 and r'_3 (ellipse);
- $E_4 = \min V_{\text{eff}}$: orbit of constant radius r_4 (circle), bounded and closed.

3.4 The Virial Theorem

The virial theorem is **statistical** in nature. Consider the quantity

$$G = \sum_i \vec{p}_i \cdot \vec{r}_i.$$

Then

$$\frac{dG}{dt} = \sum_i \dot{\vec{r}}_i \cdot \vec{p}_i + \sum_i \dot{\vec{p}}_i \cdot \vec{r}_i = \sum_i m_i \dot{\vec{r}}_i^2 + \sum_i \vec{F}_i \cdot \vec{r}_i = 2T + \sum_i \vec{F}_i \cdot \vec{r}_i.$$

Averaging over a time interval τ ,

$$\frac{\overline{dG}}{dt} = \frac{1}{\tau} [G(\tau) - G(0)] = 2\overline{T} + \overline{\sum_i \vec{F}_i \cdot \vec{r}_i}.$$

If the coordinates and velocities remain bounded, the left-hand side vanishes as $\tau \rightarrow \infty$, and we obtain the **virial theorem**

$$\overline{T} = -\frac{1}{2} \overline{\sum_i \vec{F}_i \cdot \vec{r}_i},$$

where the right-hand side is known as the **virial of Clausius**. It can be used, for instance, to deduce the equation of state of an ideal gas.

If the forces are derivable from a potential,

$$\overline{T} = \frac{1}{2} \overline{\sum_i \nabla_i V \cdot \vec{r}_i},$$

and for a single particle, $\overline{T} = \frac{1}{2} \overline{\partial V / \partial r} r$. If the potential is a power law, $V = \alpha r^{n+1}$,

$$\overline{T} = \frac{n+1}{2} \overline{V}.$$

By Euler's theorem this holds for any homogeneous potential of degree $n+1$.

3.5 The Differential Equation for the Orbit, and Integrable Power-Law Potentials

For $V = \alpha r^{n+1}$ the orbit equation can be integrated in terms of

- trigonometric functions for $n = 1, -2, -3$;
- elliptic functions for $n = 5, 3, 0, -4, -5, -7$.

3.6 Conditions for Closed Orbits (Bertrand's Theorem)

Circular orbits under a power-law force $f = -kr^n$ are stable if $n > -3$. Writing $u = 1/r$ and considering slight deviations from a circular orbit, $u = u_0 + a \cos(\beta\theta)$, one finds

$$\beta^2 = 3 + \frac{r}{f} \frac{df}{dr} \bigg|_{r=r_0}.$$

If β is a rational number p/q , after q revolutions the orbit begins to retrace itself (it is closed). For this to hold for all nearly circular orbits,

$$f(r) = -\frac{k}{r^{3-\beta^2}}.$$

J. Bertrand found that, going beyond first-order deviations from circularity, the orbits are closed only for $\beta^2 = 1$ and $\beta^2 = 4$: the only central forces that result in closed orbits for all bound particles are the inverse-square law and Hooke's law.

3.10 Scattering in a Central Force Field

The scattering problem is concerned with the scattering of particles by a centre of force. We assume that the force falls off to zero at very large distances. The incident beam is characterised by its **intensity** (flux density) I : the number of particles crossing unit area normal to the beam per unit time.

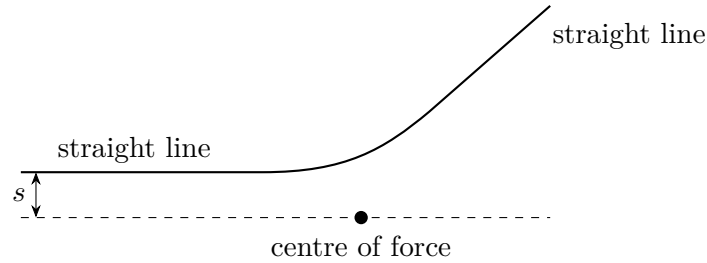


Figure 3.4: A particle with impact parameter s is deflected by the centre of force.

If the final direction differs from the initial one, the particle is said to be scattered. The **differential scattering cross section** is defined by

$$\sigma(\Omega)d\Omega := \frac{\text{number of particles scattered into solid angle } d\Omega \text{ per unit time}}{\text{incident intensity}}.$$

For central forces there is complete symmetry around the axis of the incident beam, so $d\Omega = 2\pi \sin \Theta d\Theta$.

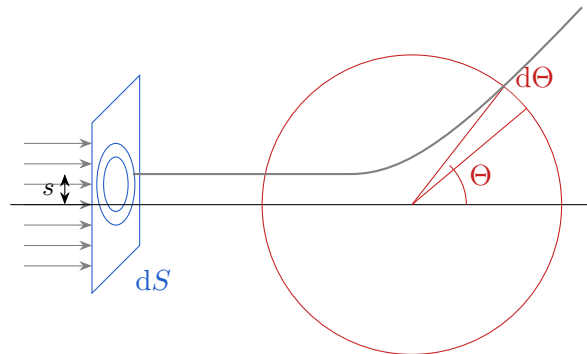


Figure 3.5: Particles crossing the ring of radius s and width ds are scattered into the angular range $d\Theta$.

For a given particle, the constants of the orbit, and hence the amount of scattering, are determined by its energy E and angular momentum l :

$$l = mv_0 s = s\sqrt{2mE}.$$

The particles with impact parameter between s and $s + ds$ are scattered into $d\Theta$:

$$2\pi I s |ds| = 2\pi \sigma(\Theta) I \sin \Theta |d\Theta| \implies \sigma(\Theta) = \frac{s}{\sin \Theta} \left| \frac{ds}{d\Theta} \right|.$$

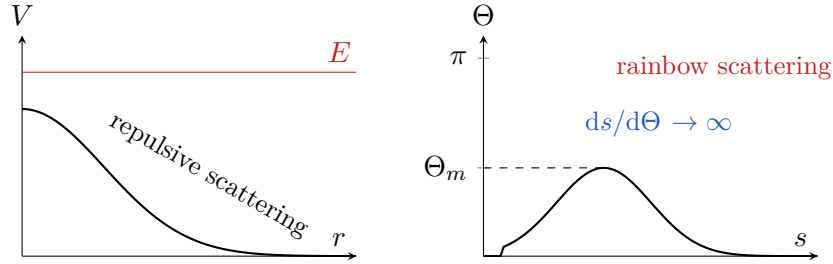


Figure 3.6: A repulsive potential with a finite maximum (left) can give a deflection function $\Theta(s)$ with a maximum Θ_m (right), where $ds/d\Theta \rightarrow \infty$: rainbow scattering.

3.11 Transformation of the Scattering Problem to Laboratory Coordinates

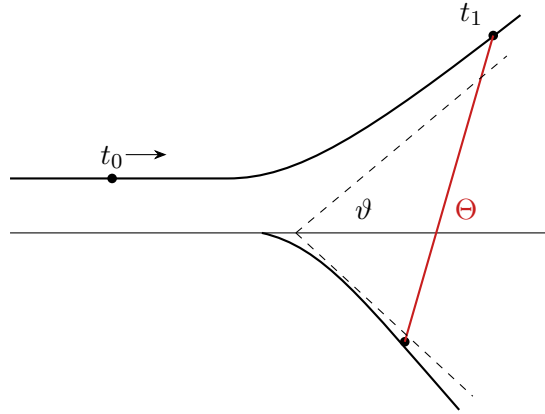


Figure 3.7: Trajectories of projectile and (initially stationary) target in the laboratory: ϑ is the laboratory scattering angle of the projectile and Θ the scattering angle of the relative vector (centre-of-mass angle).

In the laboratory system the target is initially at rest. By definition of the centre-of-mass coordinates, after scattering,

$$\vec{r}_1 = \vec{R} + \vec{r}'_1, \quad \vec{v}_1 = \vec{V} + \vec{v}'_1,$$

where $\vec{V}$ is the velocity of the centre of mass.

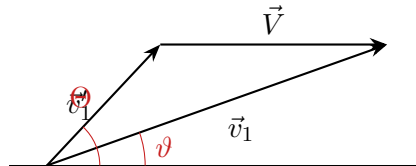


Figure 3.8: Velocity of the projectile after the collision in the laboratory ($\vec{v}_1$) and in the centre-of-mass system ($\vec{v}'_1$).

Momentum conservation, with the target at rest,

$$m_1 \vec{v}_0 = (m_1 + m_2) \vec{V} \implies \vec{V} = \frac{\mu}{m_2} \vec{v}_0, \quad \mu = \frac{m_1 m_2}{m_1 + m_2}.$$

From the figure,

$$v_1 \sin \vartheta = v'_1 \sin \Theta, \quad v_1 \cos \vartheta = v'_1 \cos \Theta + V,$$

so

$$\tan \vartheta = \frac{\sin \Theta}{\cos \Theta + \rho}, \quad \rho := \frac{V}{v'_1} = \frac{\mu}{m_2} \frac{v_0}{v'_1}.$$

Moreover, $v_1^2 = v'^2_1 + V^2 + 2v'_1 V \cos \Theta$, which gives

$$\cos \vartheta = \frac{\cos \Theta + \rho}{\sqrt{1 + 2\rho \cos \Theta + \rho^2}}.$$

By definition of the centre of mass, $v'_1 = \frac{\mu}{m_1} v$, where v is the relative speed after the collision. Hence

$$\rho = \frac{m_1}{m_2} \frac{v_0}{v};$$

if the collision is elastic, $v = v_0$ and $\rho = m_1/m_2$; if it is inelastic, $\frac{1}{2}\mu v^2 = \frac{1}{2}\mu v_0^2 + Q$.

In a given experiment the number of particles scattered into a given solid angle must be the same in both descriptions:

$$2\pi I\sigma(\Theta) \sin \Theta |d\Theta| = 2\pi I\sigma'(\vartheta) \sin \vartheta |d\vartheta| \implies \sigma'(\vartheta) = \sigma(\Theta) \left. \frac{\sin \Theta}{\sin \vartheta} \right| \frac{d\Theta}{d\vartheta},$$

that is,

$$\sigma'(\vartheta) = \sigma(\Theta) \frac{(1 + 2\rho \cos \Theta + \rho^2)^{3/2}}{|1 + \rho \cos \Theta|}.$$

4 The Kinematics of Rigid Body Motion

4.1 The Independent Coordinates of a Rigid Body

A **rigid body** is a system of mass points subject to the holonomic constraints that the distances between all pairs of points remain constant throughout the motion.

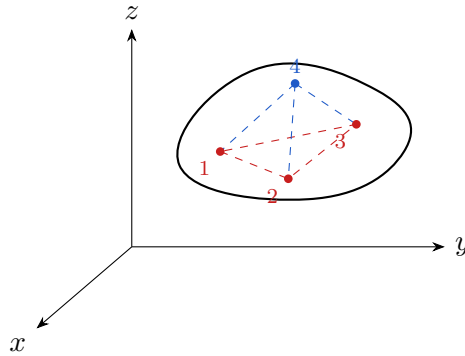


Figure 4.1: The position of any point (e.g. 4) of a rigid body is fixed by its distances to three non-collinear points 1, 2, 3.

If the body is made of N particles there are at most $3N$ degrees of freedom, but just three (non-collinear) points are needed to specify the positions of all other points. So at most 9 coordinates are needed. However, the choice of those three points is not arbitrary: the three distances between them are fixed. Point 1 needs three coordinates; point 2 lies on a sphere around 1 (two coordinates); point 3 lies on a circle about the axis 1–2 (one coordinate). Hence **just six degrees of freedom are needed**.

4.2 Orthogonal Transformations

A change of coordinate system is described by

$$x'_i = A_{ij}x_j.$$

Since the magnitude of a vector must remain unchanged,

$$x'_i x'_i = x_i x_i \implies A_{ij} A_{ik} x_j x_k = x_j x_j \implies A_{ij} A_{ik} = \delta_{jk} :$$

this is an **orthogonal transformation**.

For a rotation by ϕ in the $x_1 x_2$ plane,

$$A = \begin{pmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

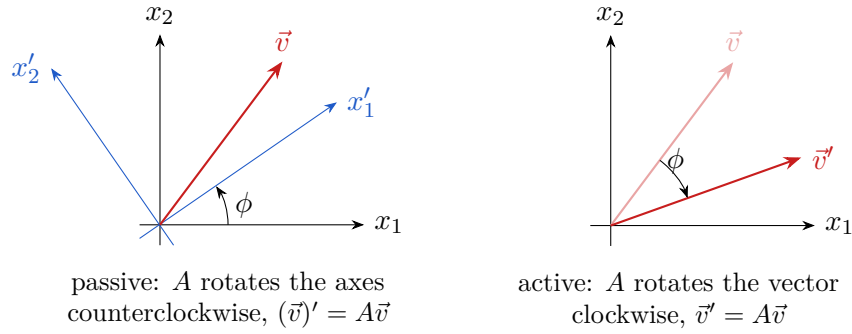


Figure 4.2: The same matrix A interpreted as a passive transformation (acting on the coordinate system) or as an active one (acting on the vector).

4.4 The Euler Angles

A real orthogonal matrix has $|A| = \pm 1$:

- $|A| = +1$: **proper** transformation;
- $|A| = -1$: **improper** transformation.

Improper transformations are not physical rigid motions. An example is an inversion: an inversion of a right-handed system into a left-handed one cannot be accomplished by any rigid change in the orientation of the coordinate axes. Another way of reaching this conclusion starts from the fact that the transformation matrix must evolve continuously from the unit matrix, which has determinant $+1$.

To describe the motion of rigid bodies in the Lagrangian formulation, three independent parameters are needed. A general convention is to use the **Euler angles**.

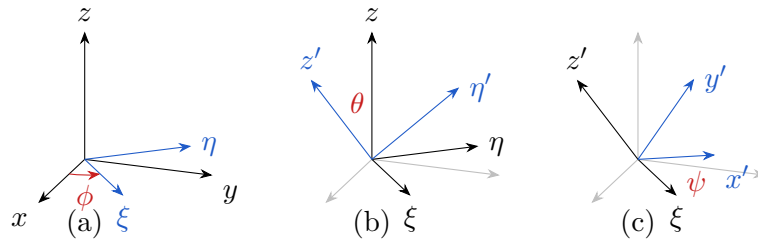


Figure 4.3: Euler angles (x -convention): (a) rotation by ϕ about z ; (b) rotation by θ about the new ξ axis (line of nodes); (c) rotation by ψ about the new z' axis.

The complete transformation is

$$A = \begin{pmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which is the ***x-convention*** of the Euler angles.

4.6 Euler's Theorem on the Motion of a Rigid Body

Statement (Euler's theorem). The general displacement of a rigid body with one point fixed is a rotation about some axis.

One direction is unaffected by the rotation: $\vec{R}' = A\vec{R} = \vec{R}$, i.e. A has the eigenvalue 1. Equivalently: *the real orthogonal matrix specifying the physical motion of a rigid body with one point fixed always has the eigenvalue +1.*

Proof. Since $AA^T = \mathbb{1}$,

$$(A - \mathbb{1})A^T = \mathbb{1} - A^T \implies |A - \mathbb{1}| \underbrace{|A^T|}_{+1} = |\mathbb{1} - A^T| = |\mathbb{1} - A|.$$

In general, for an $n \times n$ matrix B , $|-B| = (-1)^n|B|$; for $n = 3$, $|\mathbb{1} - A| = -|A - \mathbb{1}|$, so

$$|A - \mathbb{1}| = 0,$$

which is the eigenvalue equation with $\lambda = 1$. ■

Notice that $|A| = \lambda_1\lambda_2\lambda_3 = 1$ with $\lambda_3 = 1$, so $\lambda_1\lambda_2 = 1$. Since A is real, $\lambda_2 = \lambda_1^*$, hence $\lambda_{1,2} = e^{\pm i\phi}$. Taking the rotation axis along z ,

$$A = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{Tr } A = \lambda_1 + \lambda_2 + \lambda_3 = 1 + 2 \cos \phi.$$

Statement (Chasles' theorem). The most general displacement of a rigid body is a translation plus a rotation.

4.8 Infinitesimal Rotations

Write $A = \mathbb{1} + \varepsilon$, so $A^T = \mathbb{1} + \varepsilon^T$. Then

$$AA^T = (\mathbb{1} + \varepsilon)(\mathbb{1} + \varepsilon^T) = \mathbb{1} + \varepsilon + \varepsilon^T + \mathcal{O}(\varepsilon^2) = \mathbb{1} \implies \varepsilon = -\varepsilon^T:$$

if A is orthogonal, the infinitesimal matrix ε is antisymmetric,

$$\varepsilon = \begin{pmatrix} 0 & d\Omega_3 & -d\Omega_2 \\ -d\Omega_3 & 0 & d\Omega_1 \\ d\Omega_2 & -d\Omega_1 & 0 \end{pmatrix}.$$

The **infinitesimal rotation generators**

$$M_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

satisfy $[M_i, M_j] = \varepsilon_{ijk} M_k$, the Lie algebra of the rotation group. Then

$$\vec{r}' = A\vec{r} = (\mathbb{1} + \varepsilon)\vec{r}, \quad d\vec{r} = \varepsilon\vec{r}, \quad dx_i = x_j d\Omega_k - x_k d\Omega_j \quad (ijk \text{ cyclic}),$$

that is,

$$d\vec{r} = \vec{r} \times d\vec{\Omega}, \quad d\vec{\Omega} = \hat{n} d\Omega,$$

with $|d\vec{r}| = r \sin \theta d\Omega$, where θ is the angle between $\vec{r}$ and the rotation axis $\hat{n}$.

4.9 Rate of Change of a Vector

Let $\vec{G}$ be a general vector or pseudovector. Its change as seen in space is the change seen in the body plus the change due to the rotation of the body axes:

$$d\vec{G}_{\text{space}} = d\vec{G}_{\text{body}} + d\vec{G}_{\text{rot}}.$$

A vector fixed in the body system rotates with it (counterclockwise), so $d\vec{G}_{\text{rot}} = d\vec{\Omega} \times \vec{G}$. With $\vec{\omega} dt = d\vec{\Omega}$,

$$\left(\frac{d\vec{G}}{dt} \right)_{\text{space}} = \left(\frac{d\vec{G}}{dt} \right)_{\text{body}} + \vec{\omega} \times \vec{G}.$$

$\vec{\omega}$ lies along the axis of the infinitesimal rotation occurring between t and $t + dt$, a direction known as the **instantaneous axis of rotation**. In general, as an operator identity,

$$\left(\frac{d}{dt} \right)_s = \left(\frac{d}{dt} \right)_b + \vec{\omega} \times .$$

5 The Rigid Body Equations of Motion

5.1 Angular Momentum and Kinetic Energy of Motion about a Point

According to Chasles' theorem,

$$T = \underbrace{\frac{1}{2} M v^2}_{\text{translation}} + \underbrace{T'(\phi, \theta, \psi)}_{\text{rotation}}.$$

The instantaneous angular velocity $\vec{\omega}$ is independent of the choice of origin of the body system of axes. Let the origins of two body systems be at $\vec{R}_1$ and $\vec{R}_2$, with $\vec{R} = \vec{R}_2 - \vec{R}_1$ fixed in the body. Then

$$\begin{aligned} \left(\frac{d\vec{R}}{dt} \right)_s &= \left(\frac{d\vec{R}_2}{dt} \right)_s - \left(\frac{d\vec{R}_1}{dt} \right)_s = \vec{\omega}_1 \times \vec{R}, \\ \left(\frac{d\vec{R}}{dt} \right)_s &= \vec{\omega}_2 \times \vec{R}, \end{aligned}$$

using the rate-of-change formula with $\vec{R}$ fixed in either body frame ($(d\vec{R}/dt)_b = 0$). Hence $(\vec{\omega}_1 - \vec{\omega}_2) \times \vec{R} = 0$ for any $\vec{R}$, and $\vec{\omega}_1 = \vec{\omega}_2$.

The total angular momentum about a fixed point is

$$\vec{L} = \sum_i m_i \vec{r}_i \times \vec{v}_i,$$

where $\vec{r}_i$ is fixed relative to the body, so $\vec{v}_i = \vec{\omega} \times \vec{r}_i$. In components,

$$\begin{aligned} L_k &= \sum_i m_i \varepsilon_{klm} r_{i,l} v_{i,m} = \sum_i m_i \varepsilon_{klm} r_{i,l} \varepsilon_{mst} \omega_s r_{i,t} \\ &= \sum_i m_i (\delta_{ks} \delta_{lt} - \delta_{kt} \delta_{ls}) r_{i,l} r_{i,t} \omega_s = \sum_i m_i [\delta_{ks} r_i^2 - r_{i,k} r_{i,s}] \omega_s. \end{aligned}$$

Defining the **inertia tensor**

$$I_{ks} := \sum_i m_i [\delta_{ks} r_i^2 - r_{i,k} r_{i,s}],$$

we have the linear relation

$$L_k = I_{ks} \omega_s.$$

5.3 The Inertia Tensor and the Moment of Inertia

The kinetic energy of motion about a point can be written as

$$T = \sum_i \frac{1}{2} m_i v_i^2 = \sum_i \frac{1}{2} m_i \vec{v}_i \cdot (\vec{\omega} \times \vec{r}_i) = \frac{1}{2} \vec{\omega} \cdot \sum_i m_i \vec{r}_i \times \vec{v}_i = \frac{1}{2} \vec{\omega} \cdot \vec{L},$$

so that

$$T = \frac{1}{2} \vec{\omega} \cdot \overleftrightarrow{I} \cdot \vec{\omega}.$$

If $\vec{\omega} = \omega \hat{n}$, $T = \frac{1}{2} \omega^2 (\hat{n} \cdot \overleftrightarrow{I} \cdot \hat{n})$, and we define the **moment of inertia** about the axis $\hat{n}$ as

$$I := \hat{n} \cdot \overleftrightarrow{I} \cdot \hat{n}.$$

Then

$$I = n_k I_{ks} n_s = \sum_i m_i n_k [\delta_{ks} r_i^2 - r_{i,k} r_{i,s}] n_s = \sum_i m_i [r_i^2 - (\vec{r}_i \cdot \hat{n})^2],$$

where the bracket is the square of the component of $\vec{r}_i$ perpendicular to $\hat{n}$. Therefore

$$I = \sum_i m_i (\vec{r}_i \times \hat{n})^2 = \frac{1}{\omega^2} \sum_i m_i (\vec{\omega} \times \vec{r}_i)^2 = \frac{1}{\omega^2} \sum_i m_i v_i^2 \implies I = \frac{2T}{\omega^2}.$$

Parallel axis theorem.

$$I_a = I_b + MR^2,$$

where b is an axis through the centre of mass, a a parallel axis, and R the distance between them.

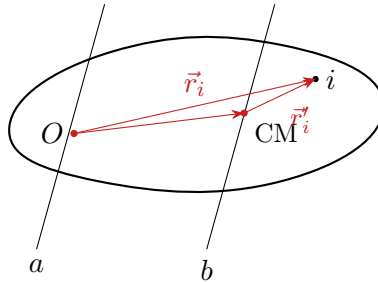


Figure 5.1: Parallel axes a (through O) and b (through the centre of mass).

5.5 Solving Rigid Body Problems and the Euler Equations of Motion

In general, for holonomic conservative systems,

$$L(q, \dot{q}) = \underbrace{L_c(q_c, \dot{q}_c)}_{\text{centre of mass}} + \underbrace{L_b(q_b, \dot{q}_b)}_{\text{orientation about the CM}}.$$

It is convenient to work in terms of the principal axes of the body, so that the kinetic energy takes a simple form. Using the Newtonian formalism,

$$\left(\frac{d\vec{L}}{dt} \right)_s = \vec{N} \implies \left(\frac{d\vec{L}}{dt} \right)_b + \vec{\omega} \times \vec{L} = \vec{N} \implies \frac{dL_i}{dt} + \varepsilon_{ijk} \omega_j L_k = N_i,$$

with $L_i = I_{ij} \omega_j$. In principal axes, $I_{ij} = I_i \delta_{ij}$, and we obtain **Euler's equations of motion for a rigid body**:

$$\begin{aligned} I_1 \dot{\omega}_1 - \omega_2 \omega_3 (I_2 - I_3) &= N_1, \\ I_2 \dot{\omega}_2 - \omega_3 \omega_1 (I_3 - I_1) &= N_2, \\ I_3 \dot{\omega}_3 - \omega_1 \omega_2 (I_1 - I_2) &= N_3. \end{aligned}$$

5.6 Torque-Free Motion of a Rigid Body

For $\vec{N} = 0$, Euler's equations become

$$I_1 \dot{\omega}_1 = \omega_2 \omega_3 (I_2 - I_3), \quad I_2 \dot{\omega}_2 = \omega_3 \omega_1 (I_3 - I_1), \quad I_3 \dot{\omega}_3 = \omega_1 \omega_2 (I_1 - I_2).$$

In this situation the kinetic energy and the angular momentum are both conserved. With $L_i = I_i \omega_i$ (principal axes x, y, z),

$$T = \frac{1}{2} I_i \omega_i^2 = \frac{L_x^2}{2I_x} + \frac{L_y^2}{2I_y} + \frac{L_z^2}{2I_z}, \quad L^2 = L_x^2 + L_y^2 + L_z^2,$$

so

$$\frac{L_x^2}{2TI_x} + \frac{L_y^2}{2TI_y} + \frac{L_z^2}{2TI_z} = 1 \quad (\text{ellipsoid}), \quad \frac{L_x^2}{L^2} + \frac{L_y^2}{L^2} + \frac{L_z^2}{L^2} = 1 \quad (\text{sphere}).$$

The motion of the system (the tip of $\vec{L}$ in the body frame) must occur on the intersection of both surfaces.

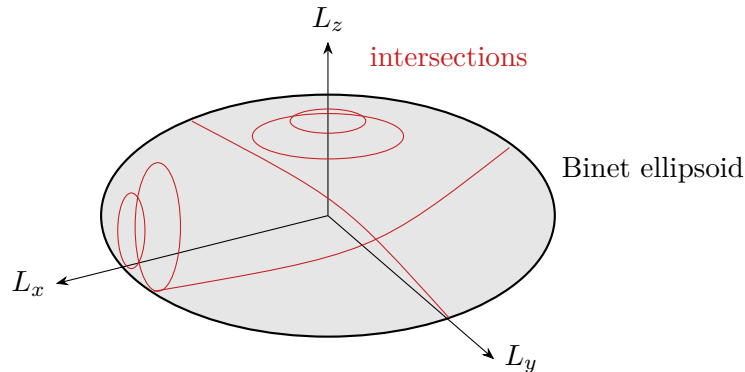


Figure 5.2: Intersections of the energy ellipsoid with spheres $L = \text{const}$, for $I_x > I_y > I_z$. The curves close around the axes of maximum and minimum moment of inertia and cross at the intermediate axis.

For $I_x > I_y > I_z$, rotation is **stable** about the axes of minimum and maximum moment of inertia, and **unstable** about the axis of intermediate moment of inertia.

Symmetric body. Let us solve the equations for $I = I_1 = I_2 \neq I_3$:

$$I\dot{\omega}_1 = (I - I_3)\omega_2\omega_3, \quad I\dot{\omega}_2 = -(I - I_3)\omega_1\omega_3, \quad I_3\dot{\omega}_3 = 0,$$

so ω_3 is constant. Defining

$$\Omega := \frac{I - I_3}{I} \omega_3,$$

we get $\dot{\omega}_1 = \Omega\omega_2$, $\dot{\omega}_2 = -\Omega\omega_1$, hence $\ddot{\omega}_1 = -\Omega^2\omega_1$ and

$$\omega_1(t) = A \sin(\Omega t), \quad \omega_2(t) = A \cos(\Omega t).$$

The component of $\vec{\omega}$ perpendicular to the symmetry axis rotates with angular frequency Ω about it. A and ω_3 (or Ω) can be written in terms of L and E or of the initial conditions:

$$T = \frac{1}{2}IA^2 + \frac{1}{2}I_3\omega_3^2, \quad L^2 = I^2A^2 + I_3^2\omega_3^2.$$

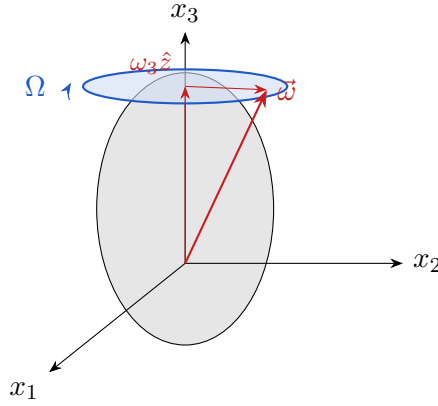


Figure 5.3: Torque-free symmetric body: $\vec{\omega}$ precesses about the symmetry axis x_3 with angular frequency Ω .

5.7 The Heavy Symmetrical Top with One Point Fixed

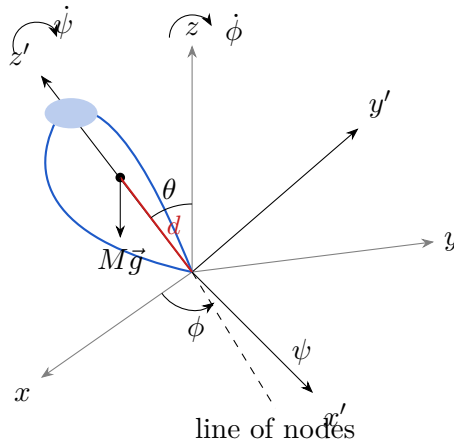


Figure 5.4: Heavy symmetric top with fixed point at the origin; the centre of gravity is at distance d along the symmetry axis z' .

$$T = \frac{1}{2}I_1(\omega_1^2 + \omega_2^2) + \frac{1}{2}I_3\omega_3^2.$$

The body-frame components of the three Euler angular velocities are

$$\begin{aligned}(\omega_\phi)_{x'} &= \dot{\phi} \sin \theta \sin \psi, & (\omega_\theta)_{x'} &= \dot{\theta} \cos \psi, & (\omega_\psi)_{x'} &= 0, \\(\omega_\phi)_{y'} &= \dot{\phi} \sin \theta \cos \psi, & (\omega_\theta)_{y'} &= -\dot{\theta} \sin \psi, & (\omega_\psi)_{y'} &= 0, \\(\omega_\phi)_{z'} &= \dot{\phi} \cos \theta, & (\omega_\theta)_{z'} &= 0, & (\omega_\psi)_{z'} &= \dot{\psi},\end{aligned}$$

so that

$$\omega_1 = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi, \quad \omega_2 = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi, \quad \omega_3 = \dot{\psi} + \dot{\phi} \cos \theta,$$

with $\omega_1^2 + \omega_2^2 = \dot{\phi}^2 \sin^2 \theta + \dot{\theta}^2$. Hence

$$T = \frac{1}{2} I_1 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} I_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2.$$

The potential energy is

$$V = - \sum_i m_i \vec{r}_i \cdot \vec{g} = -M \vec{R} \cdot \vec{g} = Mgd \cos \theta,$$

and the Lagrangian

$$L = \frac{1}{2} I_1 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} I_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2 - Mgd \cos \theta.$$

There are two cyclic coordinates, ψ and ϕ :

$$\begin{aligned}p_\psi &= \frac{\partial L}{\partial \dot{\psi}} = I_3 (\dot{\psi} + \dot{\phi} \cos \theta) = I_3 \omega_3 =: I_1 a, \\p_\phi &= \frac{\partial L}{\partial \dot{\phi}} = I_1 \dot{\phi} \sin^2 \theta + I_3 (\dot{\psi} + \dot{\phi} \cos \theta) \cos \theta =: I_1 b,\end{aligned}$$

which are conserved. The energy is also conserved:

$$E = \frac{1}{2} I_1 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} I_3 \omega_3^2 + Mgd \cos \theta.$$

From p_ψ , $I_3 \dot{\psi} = I_1 a - I_3 \dot{\phi} \cos \theta$; substituting in p_ϕ ,

$$\dot{\phi} = \frac{b - a \cos \theta}{\sin^2 \theta}, \quad \dot{\psi} = \frac{I_1 a}{I_3} - \cos \theta \frac{b - a \cos \theta}{\sin^2 \theta}.$$

Only θ is left. From the energy,

$$E = \frac{I_1}{2} \dot{\theta}^2 + \underbrace{\frac{I_1}{2} \frac{(b - a \cos \theta)^2}{\sin^2 \theta} + Mgd \cos \theta}_{V_{\text{eff}}} + \frac{I_3}{2} \omega_3^2.$$

Defining

$$\begin{aligned}\alpha &:= \frac{2E - I_3 \omega_3^2}{I_1}, & \beta &:= \frac{2Mgd}{I_1}, \\ \alpha &= \dot{\theta}^2 + \frac{(b - a \cos \theta)^2}{\sin^2 \theta} + \beta \cos \theta,\end{aligned}$$

and with $u = \cos \theta$,

$$\dot{u}^2 = (1 - u^2)(\alpha - \beta u) - (b - au)^2 =: f(u), \quad t = \int_{u(0)}^{u(t)} \frac{du}{\sqrt{(1 - u^2)(\alpha - \beta u) - (b - au)^2}}.$$

It is possible to extract qualitative information from $\dot{u}^2 = f(u)$: we need $f(u) \geq 0$ and $|u| \leq 1$, and f is a cubic polynomial with $f(\pm 1) = -(b \mp a)^2 \leq 0$ and $f(u) \rightarrow +\infty$ for $u \rightarrow +\infty$.

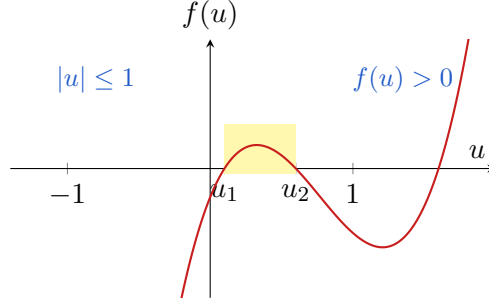


Figure 5.5: The allowed region is $u_1 \leq u \leq u_2$, where $f(u) \geq 0$ inside $|u| \leq 1$: θ moves between two angles θ_1 and θ_2 .

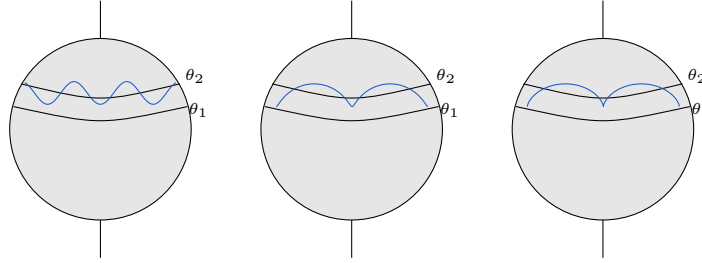


Figure 5.6: The three possible types of motion of the symmetry axis between θ_1 and θ_2 (undulating, looping, and cusped).

In this case there is not just **precession** ($\dot{\phi} \neq 0$) but also **nutation** ($\theta_1 < \theta < \theta_2$).

6 Oscillations

6.1 Formulation of the Problem

Consider conservative systems with $V = V(\vec{r})$, and assume $q_i = f(\vec{r}_i)$, not $f(\vec{r}_i, t)$ (excluding time-dependent constraints). The system is said to be in **equilibrium** when the **generalised forces acting on the system vanish**:

$$Q_i = -\frac{\partial V}{\partial q_i} = 0,$$

i.e. at a critical point of the potential (extremum or saddle point). If the configuration is initially at the equilibrium position with $\dot{q}_i = 0$, the system remains there indefinitely: since $V = V(q)$ and T is quadratic in the $\dot{q}$'s,

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} \right) = \underbrace{\frac{\partial T}{\partial q_i}}_0 - \underbrace{\frac{\partial V}{\partial q_i}}_0 \implies \frac{\partial T}{\partial \dot{q}_i} = \text{const} = 0.$$

- **Stable equilibrium**: a small disturbance of the system from equilibrium results only in small **bounded motion** about the rest position.
- **Unstable equilibrium**: an infinitesimal disturbance eventually produces **unbounded motion**.

Write the deviations of the generalised coordinates as $q_i = q_{0i} + \delta q_i$. Then

$$V = \underbrace{V(q_0)}_{\text{arbitrary}} + \underbrace{\left. \frac{\partial V}{\partial q_i} \right|_{q_0}}_{0 \text{ (equilibrium)}} \delta q_i + \underbrace{\frac{1}{2} \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{q_0}}_{V_{ij}} \delta q_i \delta q_j + \mathcal{O}(\delta q^3).$$

The kinetic energy is $T = \frac{1}{2} m_{ij} \delta \dot{q}_i \delta \dot{q}_j$, where in general $m_{ij} = m_{ij}(q)$; to lowest order

$$T = \frac{1}{2} (m_{ij}(q_0) + \mathcal{O}(\delta q)) \delta \dot{q}_i \delta \dot{q}_j \simeq \frac{1}{2} T_{ij} \delta \dot{q}_i \delta \dot{q}_j, \quad T_{ij} := m_{ij}(q_0).$$

Thus the Lagrangian for the perturbations is

$$L = \frac{1}{2} (T_{ij} \delta \dot{q}_i \delta \dot{q}_j - V_{ij} \delta q_i \delta q_j).$$

With

$$\frac{\partial L}{\partial \delta \dot{q}_k} = T_{kj} \delta \dot{q}_j, \quad \frac{\partial L}{\partial \delta q_k} = -V_{kj} \delta q_j,$$

the equations of motion for the perturbations are

$$T_{ij} \delta \ddot{q}_j + V_{ij} \delta q_j = 0. \quad (6.1)$$

In most cases of interest T_{ij} is diagonal, $T_{ij} = T_i \delta_{ij}$, and

$$L = \frac{1}{2} (T_i \delta \dot{q}_i^2 - V_{ij} \delta q_i \delta q_j), \quad \delta_{ij} T_i \delta \ddot{q}_j + V_{ij} \delta q_j = 0.$$

6.2 The Eigenvalue Equation and the Principal Axis Transformation

Equation (6.1) can be solved with the ansatz $\delta q_k = C a_k e^{-i\omega t}$, which yields

$$V_{jk} a_k - \omega^2 T_{jk} a_k = 0 \quad \implies \quad (V_{jk} - \omega^2 T_{jk}) a_k = 0.$$

Non-trivial solutions appear if we solve the eigenvalue problem

$$\det(V_{jk} - \omega^2 T_{jk}) = 0.$$

This gives n eigenvalues $\omega_{(k)}^2$ and n eigenvectors $\alpha^{(k)}$ with

$$(V_{ij} - \omega_{(k)}^2 T_{ij}) \alpha_j^{(k)} = 0.$$

For each k only $n - 1$ of these equations are independent, so we have the freedom to impose an additional condition (a normalisation). Later, δq_i can be written in terms of the $\alpha_j^{(k)}$.

Notice that

$$V_{ij} \alpha_j^{(k)} = \omega_{(k)}^2 T_{ij} \alpha_j^{(k)}, \quad V_{ij} \alpha_j^{(l)*} = \omega_{(l)}^{2*} T_{ij} \alpha_j^{(l)*}$$

(the second being the complex conjugate, with l in place of k). Multiplying by the other eigenvector,

$$V_{ij} \alpha_j^{(k)} \alpha_i^{(l)*} = \omega_{(k)}^2 T_{ij} \alpha_j^{(k)} \alpha_i^{(l)*}, \quad V_{ij} \alpha_j^{(l)*} \alpha_i^{(k)} = \omega_{(l)}^{2*} T_{ij} \alpha_j^{(l)*} \alpha_i^{(k)},$$

and subtracting (using the symmetry of V_{ij} and T_{ij}),

$$\underbrace{(V_{ij} - V_{ji})}_{0} \alpha_j^{(k)} \alpha_i^{(l)*} = (\omega_{(k)}^2 - \omega_{(l)}^{2*}) T_{ij} \alpha_j^{(k)} \alpha_i^{(l)*} \implies 0 = (\omega_{(k)}^2 - \omega_{(l)}^{2*}) T_{ij} \alpha_j^{(k)} \alpha_i^{(l)*}.$$

- If $k \neq l$ and $\omega_{(k)}^2 \neq \omega_{(l)}^2$ (no degeneracy), then $\alpha_i^{(l)*} T_{ij} \alpha_j^{(k)} = 0$.
- If $k = l$, then $0 = 2i \operatorname{Im}(\omega_{(k)}^2) T_{ij} \alpha_i^{(k)*} \alpha_j^{(k)}$, and since $T_{ij} \alpha_i^{(k)*} \alpha_j^{(k)}$ is real and non-zero (T is positive definite), $\omega_{(k)}^2$ is real.

We still have the freedom to impose a condition on $\alpha^{(k)}$; we choose $T_{ij} \alpha_i^{(k)*} \alpha_j^{(k)} = 1$, so that

$$\alpha_i^{(l)*} T_{ij} \alpha_j^{(m)} = \delta_{lm}.$$

Multiplying $V_{ij} \alpha_j^{(k)} - \omega_{(k)}^2 T_{ij} \alpha_j^{(k)} = 0$ by $\alpha_i^{(l)}$ (real eigenvectors can be chosen),

$$\alpha_i^{(l)} V_{ij} \alpha_j^{(k)} = \omega_{(k)}^2 \delta_{lk} :$$

T_{ij} and V_{ij} are diagonalised simultaneously by the matrix $\alpha_{ik} := \alpha_i^{(k)}$.

6.3 Frequencies of Free Vibration, and Normal Coordinates

The general solution can be written as

$$\delta q_i = C_k a_{ik} e^{-i\omega_k t},$$

with complex scale factors C_k , eigenvectors a_{ik} and resonant frequencies ω_k . Since ω_k and $-\omega_k$ are both solutions, explicitly

$$\delta q_i = a_{ik} (C_k^+ e^{i\omega_k t} + C_k^- e^{-i\omega_k t}).$$

The physical solution corresponds to the real part,

$$\delta q_i = f_k a_{ik} \cos(\omega_k t + \delta_k),$$

where the amplitudes f_k and phases δ_k (like $C_k^\pm$) are determined by the initial conditions.

It is possible to transform the δq_i to a new set of generalised coordinates that are **all simple periodic functions of time**, known as **normal coordinates**. Write

$$\delta q_i = \alpha_{ik} \zeta_k,$$

where the ζ_k are the normal coordinates and α_{ik} the diagonalising matrix. Then

$$L = \frac{1}{2} (T_{ij} \delta \dot{q}_i \delta \dot{q}_j - V_{ij} \delta q_i \delta q_j) = \frac{1}{2} \left(\underbrace{T_{ij} \alpha_{ik} \alpha_{jl}}_{\delta_{kl}} \dot{\zeta}_k \dot{\zeta}_l - \underbrace{V_{ij} \alpha_{ik} \alpha_{jl}}_{\omega_{(k)}^2 \delta_{kl}} \zeta_k \zeta_l \right) = \frac{1}{2} (\dot{\zeta}_k^2 - \omega_k^2 \zeta_k^2),$$

so that

$$\ddot{\zeta}_k + \omega_k^2 \zeta_k = 0, \quad \zeta_k = C_k e^{-i\omega_k t}.$$

Each normal coordinate corresponds to a **vibration of the system with only one frequency**. These oscillations are known as **normal modes**.

8 The Hamilton Equations of Motion

In the following chapters the systems are assumed to be holonomic and monogenic, with $V = V(q, \dot{q})$ and not $V(q, \dot{q}, t)$.

8.1 Legendre Transformations and the Hamilton Equations of Motion

In the Hamiltonian formalism there can be no constraint equations (n independent coordinates). The Hamiltonian formalism uses first-order equations of motion: $2n$ of them describe the behaviour in phase space, in terms of n generalised coordinates q_i and n conjugate momenta $p_i = \partial L / \partial \dot{q}_i$. The q and p are called **canonical variables**. One passes from the Lagrangian $(q, \dot{q}, t)$ to the Hamiltonian (q, p, t) by a **Legendre transformation**. With $L = L(q, \dot{q}, t)$,

$$dL = \underbrace{\frac{\partial L}{\partial q_i} dq_i}_{\dot{p}_i} + \underbrace{\frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i}_{p_i} + \frac{\partial L}{\partial t} dt.$$

Define

$$H(q, p, t) := \dot{q}_i p_i - L.$$

Then

$$\begin{aligned} dH &= \frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial t} dt = \dot{q}_i dp_i + p_i d\dot{q}_i - \left(\dot{p}_i dq_i + p_i d\dot{q}_i + \frac{\partial L}{\partial t} dt \right) \\ &= \dot{q}_i dp_i - \dot{p}_i dq_i - \frac{\partial L}{\partial t} dt, \end{aligned}$$

which gives the **canonical equations of Hamilton**

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad \dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \frac{\partial L}{\partial t} = -\frac{\partial H}{\partial t}.$$

The canonical equations are not symmetric, but they can be conveniently arranged to look that way. For a system with n degrees of freedom define

$$\eta_i := q_i, \quad \eta_{i+n} := p_i, \quad i \leq n,$$

so that $\partial H / \partial \eta_i = \partial H / \partial q_i$ and $\partial H / \partial \eta_{i+n} = \partial H / \partial p_i$, and

$$J = \begin{pmatrix} 0_{n \times n} & \mathbb{1}_{n \times n} \\ -\mathbb{1}_{n \times n} & 0_{n \times n} \end{pmatrix}_{2n \times 2n}, \quad J^T = -J, \quad J^{-1} = J^T = -J, \quad J^2 = -\mathbb{1}, \quad \det J = +1.$$

Then Hamilton's equations can be written compactly ("intertwined") as

$$\dot{\eta}_i = J_{ik} \frac{\partial H}{\partial \eta_k},$$

which is **Hamilton's equations in symplectic notation**.

8.2 Cyclic Coordinates and Conservation Theorems

If L does not depend on q_i , neither does H , so the momentum conservation theorem carries over from L to H : since $\dot{p}_i = -\partial H / \partial q_i$, if $H \neq H(q_i)$ then $\dot{p}_i = 0$ and p_i is conserved. Moreover,

$$\frac{dH}{dt} = \frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i + \frac{\partial H}{\partial t} = -\dot{p}_i \dot{q}_i + \dot{q}_i \dot{p}_i + \frac{\partial H}{\partial t} \implies \frac{dH}{dt} = \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.$$

If the **Lagrangian does not depend explicitly on time**, then the **Hamiltonian** does not either, and **it is conserved**. If $\vec{r}_i = \vec{r}_i(q_j)$ (no explicit t) and $V = V(q)$ (no dependence on $\dot{q}$), then $H = T + V$ is the total energy and is conserved.

Using a different set of generalised coordinates in the definition of the Hamiltonian may lead to an entirely different quantity. **It may be that for one set of generalised coordinates H is conserved, but that for another it varies in time.**

Example. A mass m is attached by a spring of constant k to a cart moving with constant velocity v_0 .

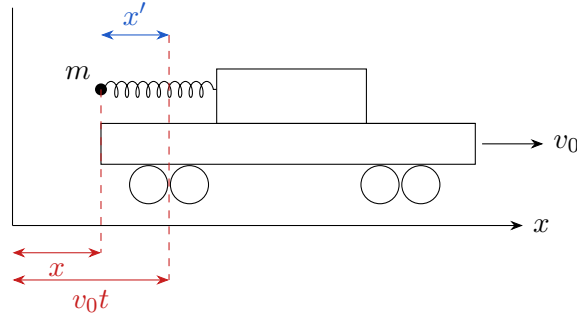


Figure 8.1: Mass attached by a spring to a cart moving with constant velocity v_0 .

x coordinate.

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}k(x - v_0 t)^2, \quad p = m\dot{x}, \quad H = \frac{p^2}{2m} + \frac{1}{2}k(x - v_0 t)^2.$$

Since $L = L(x, \dot{x}, t)$, H is the energy, but it is not conserved. The equation of motion is $m\ddot{x} = -k(x - v_0 t)$; with $y = x - v_0 t$, $m\ddot{y} = -ky$.

x' coordinate. With $x' = x - v_0 t$, i.e. $x = x' + v_0 t$,

$$L = \frac{1}{2}m(\dot{x}' + v_0)^2 - \frac{1}{2}kx'^2, \quad p' = m(\dot{x}' + v_0) \implies \dot{x}' = \frac{p'}{m} - v_0,$$

$$H = p'\dot{x}' - L = p'\left(\frac{p'}{m} - v_0\right) - \frac{1}{2}\frac{p'^2}{m} + \frac{1}{2}kx'^2 = \frac{1}{2m}(p' - mv_0)^2 + \frac{1}{2}kx'^2 - \frac{1}{2}mv_0^2.$$

H is not the energy, but it is conserved. The equation of motion is $m\ddot{x}' = -kx'$: both descriptions give the same equations of motion.

9 Canonical Transformations

9.1 The Equations of Canonical Transformations

There is one type of problem for which the solution of Hamilton's equations is trivial: when all coordinates are cyclic. Then $p_i = \alpha_i$ (constants), $H = H(\alpha_i)$, and

$$\dot{q}_i = \frac{\partial H}{\partial \alpha_i} = \omega_i(\alpha) \implies q_i = \omega_i t + \beta_i,$$

where the β_i are constants of integration. We must find the procedure for transforming one set of variables into another that may be more suitable:

$$Q_i = Q_i(q, p, t), \quad P_i = P_i(q, p, t),$$

point transformations in phase space. The transformations of interest are those for which the new coordinates are also canonical,

$$\dot{Q}_i = \frac{\partial K}{\partial P_i}, \quad \dot{P}_i = -\frac{\partial K}{\partial Q_i},$$

where K plays the role of the Hamiltonian in the new set. The new quantities must satisfy a modified Hamilton's principle,

$$\delta \int_{t_1}^{t_2} (P_i \dot{Q}_i - K(Q, P, t)) dt = 0,$$

while $\delta \int_{t_1}^{t_2} (p_i \dot{q}_i - H(q, p, t)) dt = 0$ still holds. Therefore

$$\lambda(p_i \dot{q}_i - H) = P_i \dot{Q}_i - K + \frac{dF}{dt},$$

which defines an **extended canonical transformation**; λ is a scalar (a scale transformation). With the aid of a suitable scale transformation it is always possible to restrict attention to $\lambda = 1$:

$$p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{dF}{dt},$$

the **canonical transformation**. Transformations for which $Q = Q(q, p)$ and $P = P(q, p)$ (no explicit t) are called **restricted canonical transformations**. The function $F = F(q, p, Q, P, t)$ is called the **generating function**: it acts as the bridge between the two sets of variables.

Example. Take $F = F_1(q, Q, t)$. Then

$$p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{dF_1}{dt} = P_i \dot{Q}_i - K + \frac{\partial F_1}{\partial t} + \frac{\partial F_1}{\partial q_i} \dot{q}_i + \frac{\partial F_1}{\partial Q_i} \dot{Q}_i,$$

and since q and Q are independent,

$$p_i = \frac{\partial F_1}{\partial q_i}, \quad P_i = -\frac{\partial F_1}{\partial Q_i}, \quad K = H + \frac{\partial F_1}{\partial t}.$$

(The process can be reversed to find F_1 .) In general,

$$\frac{dF}{dt} = (K - H) + p_i \dot{q}_i - P_i \dot{Q}_i,$$

and the four basic types of generating functions are:

Generating function	Derivatives	Trivial special case
$F = F_1(q, Q, t)$	$p_i = \frac{\partial F_1}{\partial q_i}, \quad P_i = -\frac{\partial F_1}{\partial Q_i}$	$F_1 = q_i Q_i: Q_i = p_i, P_i = -q_i$
$F = F_2(q, P, t) - Q_i P_i$	$p_i = \frac{\partial F_2}{\partial q_i}, \quad Q_i = \frac{\partial F_2}{\partial P_i}$	$F_2 = q_i P_i: Q_i = q_i, P_i = p_i$
$F = F_3(p, Q, t) + q_i p_i$	$q_i = -\frac{\partial F_3}{\partial p_i}, \quad P_i = -\frac{\partial F_3}{\partial Q_i}$	$F_3 = p_i Q_i: Q_i = -q_i, P_i = -p_i$
$F = F_4(p, P, t) + q_i p_i - Q_i P_i$	$q_i = -\frac{\partial F_4}{\partial p_i}, \quad Q_i = \frac{\partial F_4}{\partial P_i}$	$F_4 = p_i P_i: Q_i = p_i, P_i = -q_i$

9.3 The Harmonic Oscillator

$$H = \frac{p^2}{2m} + \frac{kq^2}{2} = \frac{1}{2m}(p^2 + m^2 \omega^2 q^2), \quad \omega^2 := \frac{k}{m}.$$

Try the transformation

$$p = f(P) \cos Q, \quad q = \frac{f(P)}{m\omega} \sin Q \implies K = H = \frac{f^2(P)}{2m},$$

so that Q is cyclic. Is it canonical? In general, if $F_1(q, Q) = g(q)h(Q)$, then $p = \partial F_1/\partial q$ and $P = -\partial F_1/\partial Q$. Dividing the two relations,

$$p = m\omega q \cot Q,$$

which is obtained from

$$F_1 = \frac{m\omega q^2}{2} \cot Q,$$

and then, using $d \cot Q/dQ = -1/\sin^2 Q$,

$$P = -\frac{\partial F_1}{\partial Q} = \frac{m\omega q^2}{2 \sin^2 Q}.$$

So F_1 exists and the transformation is canonical, with

$$f(P) = \sqrt{2m\omega P}, \quad p = \sqrt{2m\omega P} \cos Q, \quad q = \sqrt{\frac{2P}{m\omega}} \sin Q.$$

Then

$$K = H = E = \omega P.$$

The equations of motion are

$$\dot{Q} = \frac{\partial K}{\partial P} = \omega, \quad \dot{P} = -\frac{\partial K}{\partial Q} = 0, \quad \Rightarrow \quad P = \frac{E}{\omega}, \quad Q = \omega t + \alpha.$$

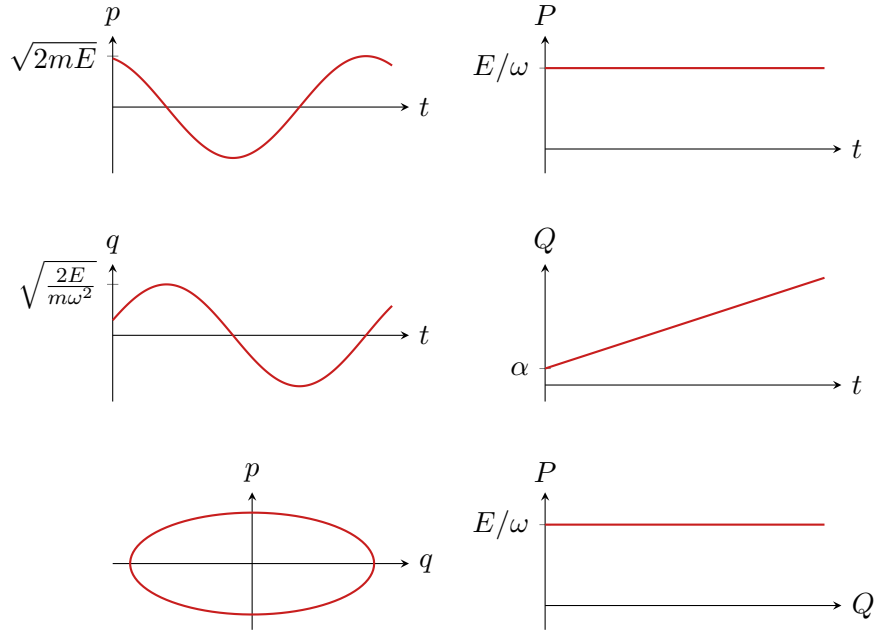


Figure 9.1: Harmonic oscillator in the original variables (q, p) (left) and in the new canonical variables (Q, P) (right).

10 Hamilton–Jacobi Theory and Action–Angle Variables

The goal is to find a transformation to a new set of constant quantities, which might be the $2n$ initial values:

$$q = q(q_0, p_0, t), \quad p = p(q_0, p_0, t).$$

This is more general than what was explained in the last chapter.

10.1 The Hamilton–Jacobi Equation for Hamilton’s Principal Function

We can ensure that the new variables are constant in time by requiring the transformed Hamiltonian K to be identically zero. Then

$$\frac{\partial K}{\partial P_i} = \dot{Q}_i = 0, \quad -\frac{\partial K}{\partial Q_i} = \dot{P}_i = 0.$$

Since $K = H + \partial F/\partial t$,

$$H(q, p, t) + \frac{\partial F}{\partial t} = 0.$$

It is convenient to take $F = F_2(q, P, t)$, so that $p_i = \partial F_2/\partial q_i$. Therefore

$$H\left(q_1, \dots, q_n; \frac{\partial F_2}{\partial q_1}, \dots, \frac{\partial F_2}{\partial q_n}; t\right) + \frac{\partial F_2}{\partial t} = 0,$$

the **Hamilton–Jacobi equation**, a partial differential equation in the $n+1$ variables $q_1, \dots, q_n, t$. F_2 is denoted by S and called **Hamilton’s principal function**. Suppose there exists a solution of the form

$$F_2 =: S = S(q_1, \dots, q_n; \alpha_1, \dots, \alpha_{n+1}; t),$$

with $n+1$ constants of integration (a **complete solution**). One of them is trivial, since if S is a solution so is $S + \alpha$. Hence in general

$$S = S(q_1, \dots, q_n; \alpha_1, \dots, \alpha_n; t),$$

where none of the constants is solely additive. Since the P_i are constants, we may take $P_i = \alpha_i$. From the transformation equations

$$p_i = \frac{\partial S(q, \alpha, t)}{\partial q_i}, \quad Q_i = \beta_i = \frac{\partial S(q, \alpha, t)}{\partial \alpha_i}$$

(the β_i being constants), and in general

$$q_i = q_i(\alpha, \beta, t), \quad p_i = p_i(\alpha, \beta, t).$$

Solving the Hamilton–Jacobi equation we are at the same time obtaining a solution of the mechanical problem: the $2n$ first-order canonical equations are traded for one first-order partial differential equation.

To a certain extent the choice of the α_i is arbitrary: we could just as well choose any n quantities γ_i that are independent functions of the α_i , $\gamma_i = \gamma_i(\alpha_1, \dots, \alpha_n)$, which may be more convenient as integration constants.

Further insight.

$$\frac{dS}{dt} = \underbrace{\frac{\partial S}{\partial q_i} \dot{q}_i}_{p_i \dot{q}_i} + \underbrace{\frac{\partial S}{\partial t}}_{-H} = p_i \dot{q}_i - H = L \implies S = \int dt L + \text{const},$$

which in itself is of no help, since to compute S this way we need $q(t)$.

When the Hamiltonian is independent of time,

$$S(q, \alpha, t) = W(q, \alpha) - \alpha t,$$

where $\alpha = H$ is constant and W is **Hamilton’s characteristic function**. Then

$$\frac{dW}{dt} = \frac{\partial W}{\partial q_i} \dot{q}_i = p_i \dot{q}_i, \quad p_i = \frac{\partial W}{\partial q_i},$$

so

$$W = \int p_i \dot{q}_i dt = \int p_i dq_i,$$

the **abbreviated action**.

10.2 The Harmonic Oscillator Problem as an Example of the Hamilton–Jacobi Method

$$H = \frac{1}{2m}(p^2 + m^2\omega^2q^2) = E, \quad \omega = \sqrt{\frac{k}{m}}.$$

With $p = \partial S/\partial q$,

$$\frac{1}{2m} \left[\left(\frac{\partial S}{\partial q} \right)^2 + m^2\omega^2q^2 \right] + \frac{\partial S}{\partial t} = 0.$$

Since $H = H(p, q)$, $S = W - \alpha t$ and

$$\frac{1}{2m} \left[\left(\frac{\partial W}{\partial q} \right)^2 + m^2\omega^2q^2 \right] = \alpha,$$

with $\alpha = H$. This can be integrated:

$$W = \sqrt{2m\alpha} \int dq \sqrt{1 - \frac{m\omega^2q^2}{2\alpha}}, \quad S = \sqrt{2m\alpha} \int dq \sqrt{1 - \frac{m\omega^2q^2}{2\alpha}} - \alpha t,$$

$$\beta = \frac{\partial S}{\partial \alpha} = \sqrt{\frac{m}{2\alpha}} \int \frac{dq}{\sqrt{1 - \frac{m\omega^2q^2}{2\alpha}}} - t \implies t + \beta = \frac{1}{\omega} \arcsin \left(q \sqrt{\frac{m\omega^2}{2\alpha}} \right).$$

Hence

$$q = \sqrt{\frac{2\alpha}{m\omega^2}} \sin(\omega(t + \beta)), \quad p = \frac{\partial S}{\partial q} = \frac{\partial W}{\partial q} = \sqrt{2m\alpha - m^2\omega^2q^2} = \sqrt{2m\alpha} \cos(\omega(t + \beta)).$$

To complete the story, from the initial conditions at $t = 0$,

$$2m\alpha = p_0^2 + m^2\omega^2q_0^2 \implies \alpha = E, \quad \tan(\omega\beta) = m\omega \frac{q_0}{p_0}.$$

10.3 The Hamilton–Jacobi Equation for Hamilton’s Characteristic Function

Consider a canonical transformation in which the new momenta are all constants of the motion α_i , and where α_1 in particular is the constant of motion H . If $W = W(q, P)$,

$$p_i = \frac{\partial W}{\partial q_i}, \quad Q_i = \frac{\partial W}{\partial P_i} = \frac{\partial W}{\partial \alpha_i},$$

and $H(q_i, p_i) = \alpha_1$, i.e.

$$H \left(q_i, \frac{\partial W}{\partial q_i} \right) = \alpha_1.$$

With $K = \alpha_1$, the canonical equations give

$$\dot{P}_i = -\frac{\partial K}{\partial Q_i} = 0 \implies P_i = \alpha_i, \quad \dot{Q}_i = \frac{\partial K}{\partial \alpha_i} = \delta_{1i},$$

so that

$$Q_1 = t + \beta_1 = \frac{\partial W}{\partial \alpha_1}, \quad Q_i = \beta_i = \frac{\partial W}{\partial \alpha_i}, \quad i \neq 1.$$

It is not necessary to take α_1 and the constants of integration in W as the new constant momenta. Occasionally it is desirable to use some particular set of n independent functions γ_i of the α ’s as the transformed momenta. Then

$$\dot{Q}_i = \frac{\partial K}{\partial \gamma_i} = v_i, \quad v_i = v_i(\gamma) \implies Q_i = v_i t + \beta_i.$$

Summary of the two methods.

Principal function S	Characteristic function W
$H = H(q, p, t)$ is any function of q, p, t	$H(q, p) = \text{const}$ is conserved
We seek new variables such that all Q_i, P_i are constants of motion	We seek new variables such that all P_i are constants
It suffices that $K \equiv 0$	It suffices that $K = H(P_i) = \alpha_1$ (all Q cyclic)
$\dot{Q}_i = \partial K / \partial P_i = 0, \dot{P}_i = -\partial K / \partial Q_i = 0$	$\dot{Q}_i = \partial K / \partial P_i = v_i, \dot{P}_i = -\partial K / \partial Q_i = 0$
$Q_i = \beta_i, P_i = \gamma_i$	$Q_i = v_i t + \beta_i, P_i = \gamma_i$
Generating function $S(q, P, t)$	Generating function $W(q, P)$
$H\left(q, \frac{\partial S}{\partial q}, t\right) + \frac{\partial S}{\partial t} = 0$	$H\left(q, \frac{\partial W}{\partial q}\right) - \alpha_1 = 0$
n non-trivial constants $\alpha_1, \dots, \alpha_n$	$n - 1$ non-trivial constants which, with α_1 , form $\alpha_1, \dots, \alpha_n$
$P_i = \gamma_i(\alpha_1, \dots, \alpha_n), S = S(q_i, \gamma_i, t)$	$P_i = \gamma_i(\alpha_1, \dots, \alpha_n), W = W(q_i, \gamma_i)$
$p_i = \partial S / \partial q_i$	$p_i = \partial W / \partial q_i$
$Q_i = \partial S / \partial \gamma_i = \beta_i$	$Q_i = \partial W / \partial \gamma_i = v_i(\gamma)t + \beta_i$

In particular the γ_i may be chosen to be the α_i themselves. One half of the transformation equations ($p_i = \partial S / \partial q_i$ or $p_i = \partial W / \partial q_i$) are fulfilled automatically, since they were used in constructing the Hamilton–Jacobi equation. The other half can be solved for q_i in terms of t and the $2n$ constants β_i, γ_i . The solution is completed by evaluating these $2n$ constants in terms of the initial values (q_0, p_0) . When H does not depend explicitly on time both methods are suitable, and the generating functions are related by

$$S(q, P, t) = W(q, P) - \alpha_1 t.$$

10.4 Separation of Variables in the Hamilton–Jacobi Equation

Under certain conditions it is possible to separate the variables in the Hamilton–Jacobi equation. A coordinate q_j is said to be **separable** if Hamilton’s principal function can be split into two additive parts,

$$S(q_1, \dots, q_n; \alpha_1, \dots, \alpha_n; t) = S_j(q_j; \alpha_1, \dots, \alpha_n; t) + S'(q_1, \dots, \hat{q}_j, \dots, q_n; \alpha_1, \dots, \alpha_n; t),$$

and the Hamilton–Jacobi equation can be split into two equations. The Hamilton–Jacobi equation is **completely separable** if all the coordinates are separable:

$$S = \sum_i S_i(q_i; \alpha_1, \dots, \alpha_n; t), \quad H_i\left(q_i; \frac{\partial S_i}{\partial q_i}; \alpha_1, \dots, \alpha_n; t\right) + \frac{\partial S_i}{\partial t} = 0.$$

If the Hamiltonian does not depend explicitly on time,

$$S_i(q_i; \alpha_1, \dots, \alpha_n; t) = W_i(q_i; \alpha_1, \dots, \alpha_n) - \alpha_i t, \\ H_i\left(q_i; \frac{\partial W_i}{\partial q_i}; \alpha_1, \dots, \alpha_n\right) = \alpha_i \quad (\text{no summation}),$$

where the α_i are the separation constants.

10.6 Action–Angle Variables in Systems of One Degree of Freedom

For simplicity, consider conservative systems with one degree of freedom, $H(q, p) = \alpha_1$, so that $p = p(q, \alpha_1)$ is a curve in phase space. There are two types of periodic motion:

- **Libration** (e.g. the harmonic oscillator): the orbit is closed, q and p are periodic functions of time with the same frequency; the motion takes place between two zeros of the kinetic energy.
- **Rotation**: p is a periodic function of q , with some period q_0 .

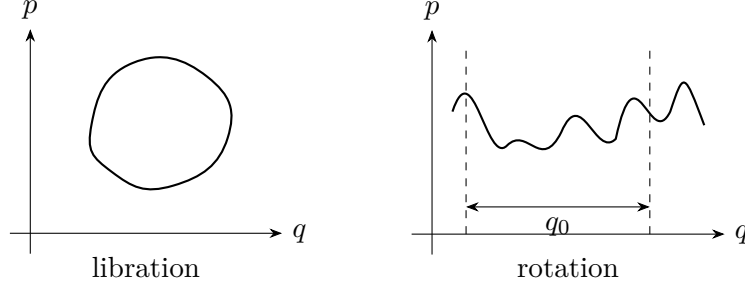


Figure 10.1: The two types of periodic motion in phase space.

Example: the pendulum.

$$E = \frac{p_\theta^2}{2ml^2} - mgl \cos \theta, \quad p_\theta = \pm \sqrt{2ml^2(E + mgl \cos \theta)}.$$

- Libration if $E < mgl$: $|\theta| \leq \theta'$, with $E = -mgl \cos \theta'$.
- Rotation if $E > mgl$: no restriction on θ .

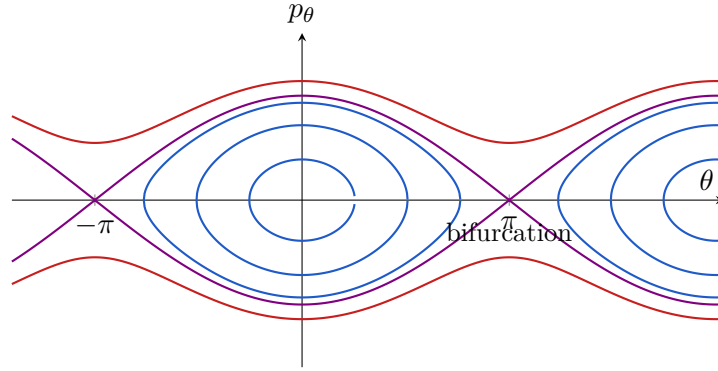


Figure 10.2: Phase portrait of the pendulum: librations (blue), the separatrix $E = mgl$ (violet), and rotations (red).

For either type of periodic motion we introduce a new variable J , designed to replace α_1 as the transformed (constant) momentum: the **action variable**

$$J := \oint p dq,$$

where the integral is over a complete period of libration or rotation. Then $\alpha_1 = H = H(J)$ and $W = W(q, J)$. The generalised **angle variable** w is defined by

$$w = \frac{\partial W}{\partial J}, \quad \dot{w} = \frac{\partial H}{\partial J} = \nu(J) \implies w = \nu t + \beta.$$

The merit of these coordinates arises from the physical interpretation of ν . Over one period,

$$\Delta w = \oint \frac{\partial w}{\partial q} dq = \oint \frac{\partial^2 W}{\partial q \partial J} dq = \frac{d}{dJ} \oint \frac{\partial W}{\partial q} dq = \frac{d}{dJ} \underbrace{\oint p dq}_J = 1 :$$

in each period w changes by unity. Since $\Delta w = 1 = \nu\tau$, $\nu = 1/\tau$ is the reciprocal of the period, i.e. the frequency associated with the periodic motion of q . J has dimensions of angular momentum, so w is an angle. The frequency of the system can be found without solving the problem completely.

Harmonic oscillator.

$$J = \oint p dq = \oint \sqrt{2m\alpha - m^2\omega^2 q^2} dq,$$

where α is the constant total energy and $\omega^2 = k/m$. The substitution $q = \sqrt{\frac{2\alpha}{m\omega^2}} \sin \theta$ reduces the integral to

$$J = \frac{2\alpha}{\omega} \int_0^{2\pi} \cos^2 \theta d\theta = \frac{2\pi\alpha}{\omega} \implies \alpha = H = \frac{J\omega}{2\pi},$$

$$\frac{\partial H}{\partial J} = \nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}},$$

the customary formula for the frequency of a linear harmonic oscillator. In terms of J and w the solution is

$$q = \sqrt{\frac{J}{\pi m \omega}} \sin(2\pi w), \quad p = \sqrt{\frac{m J \omega}{\pi}} \cos(2\pi w).$$